

An example of complex STS: air traffic control towers

Human Systems Integration – ESTIA 3A

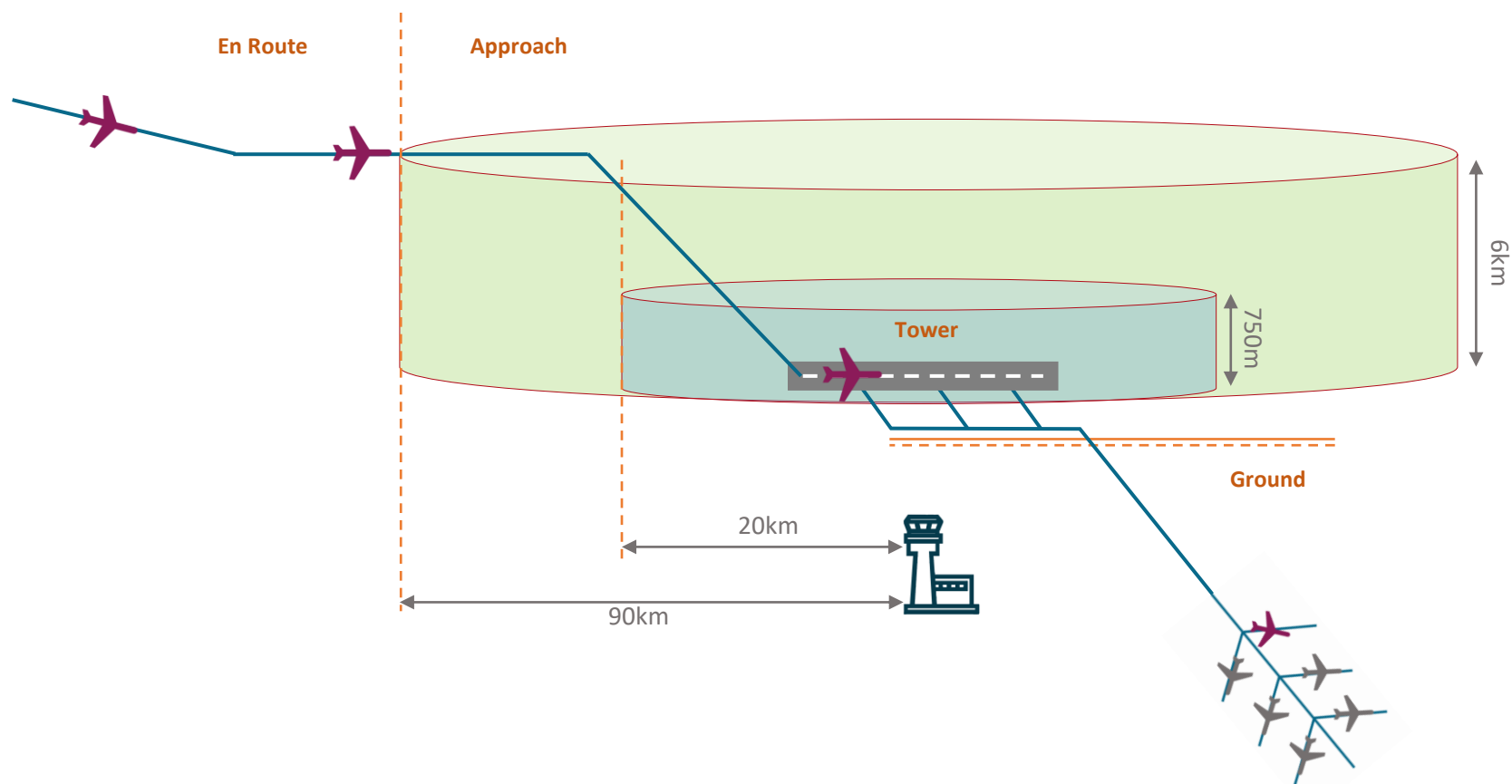
2026

Alexandre Disdier

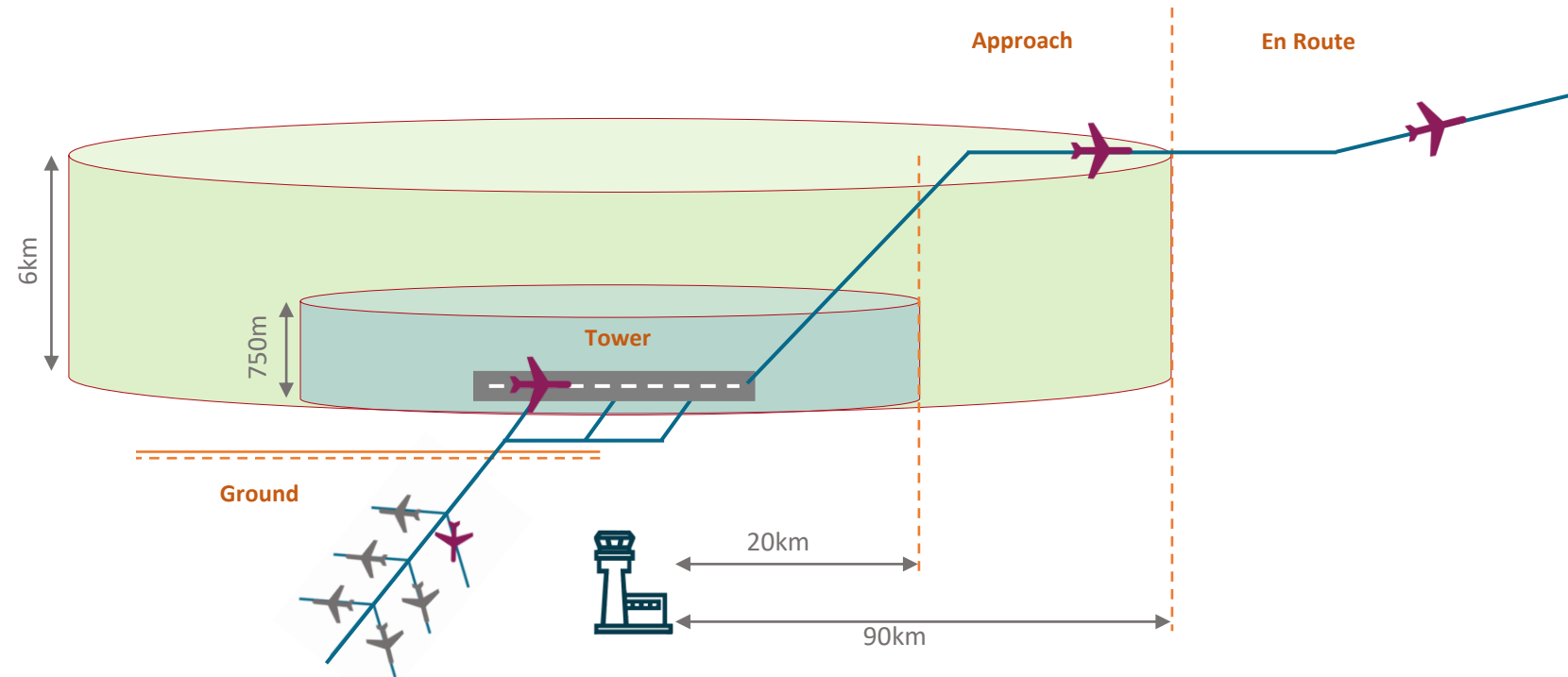
01

Motivation: remote air traffic control towers

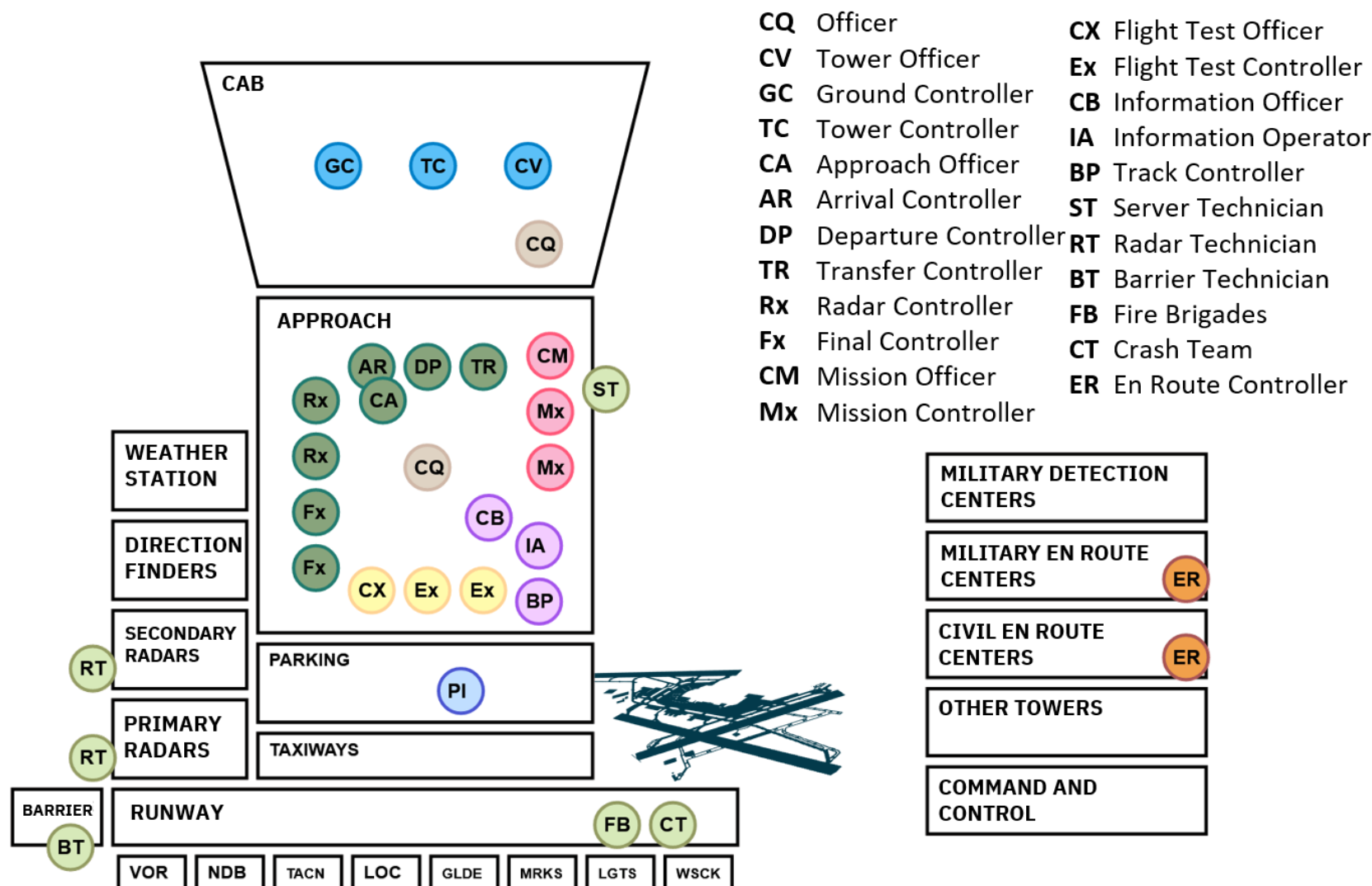
How air traffic control works: arrivals



How air traffic control works: departures

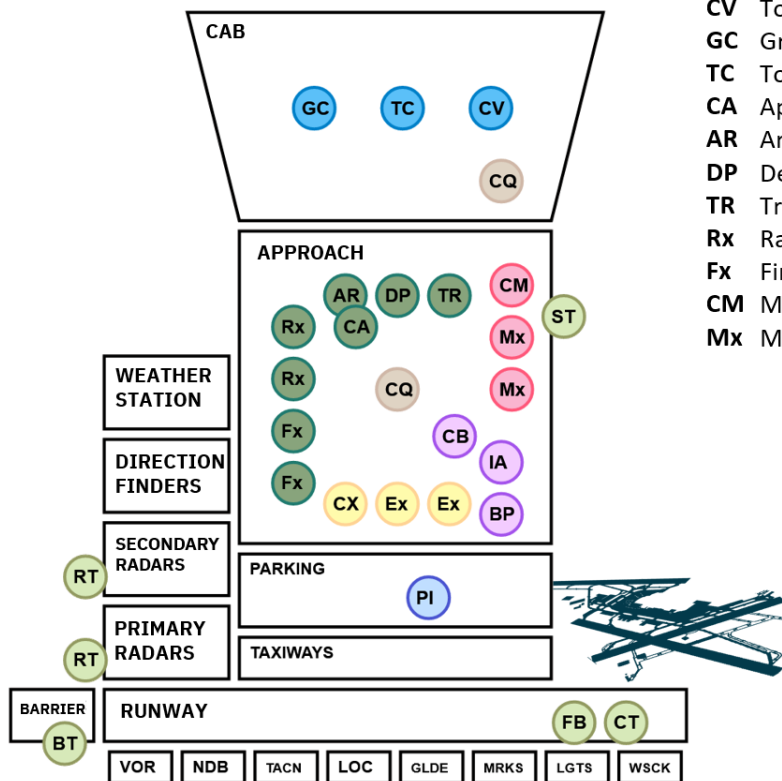


Traditional air traffic control tower



Designing remote air traffic control centers

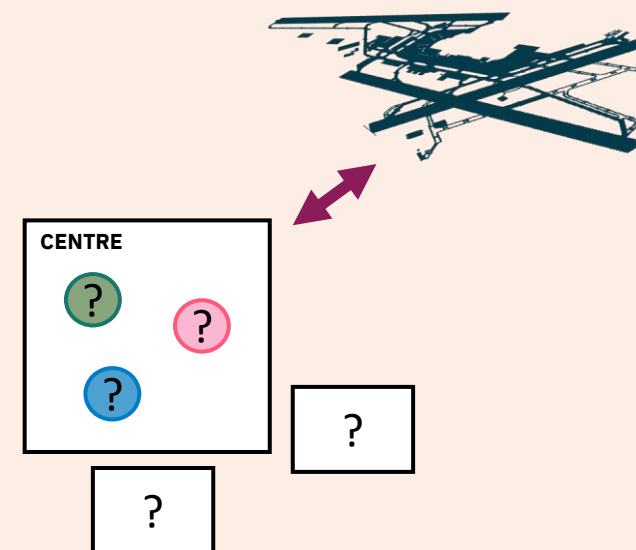
TOUR TRADITIONNELLE



CQ Officer
CV Tower Officer
GC Ground Controller
TC Tower Controller
CA Approach Officer
AR Arrival Controller
DP Departure Controller
TR Transfer Controller
Rx Radar Controller
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RT Radar Technician
BT Barrier Technician
FB Fire Brigades
CT Crash Team
ER En Route Controller

MILITARY DETECTION CENTERS
MILITARY EN ROUTE CENTERS
CIVIL EN ROUTE CENTERS
OTHER TOWERS
COMMAND AND CONTROL

REMOTE CENTER



Motivations:

Monitoring of isolated fields
 Cost reduction
 Facilitated deployment in OPEXs
 (Fürstenau, 2014)

Remote ATC Center: Why?

COST SAVINGS

Tower construction

About 40 000\$ per height feet (FAA¹)

Charlotte-Douglas Tower: 94 millions \$ (FAA²)

Tower maintenance

EASE OF DEPLOYMENT

Military context

COSCA-D : 4-hour deployment

Enemy exposure

Civilian context

Isolated terrains



RESOURCE POOLING

Human resource pooling

A single center for several low-traffic airfields

We talk about Multiple Remote Tower Operations (MRTO)

Mainly for small airfields with few airplanes per unit time

EXPLORATION OF ALTERNATIVE PROCEDURES

No tower cab means we can imagine new...

...interactions

...ways to ensure Situation Awareness (SA)

Transmission of non-visual cues (sounds, vibrations...) (Reynal et al. 2019)

Augmented Reality, Virtual Reality, Wearables, AI

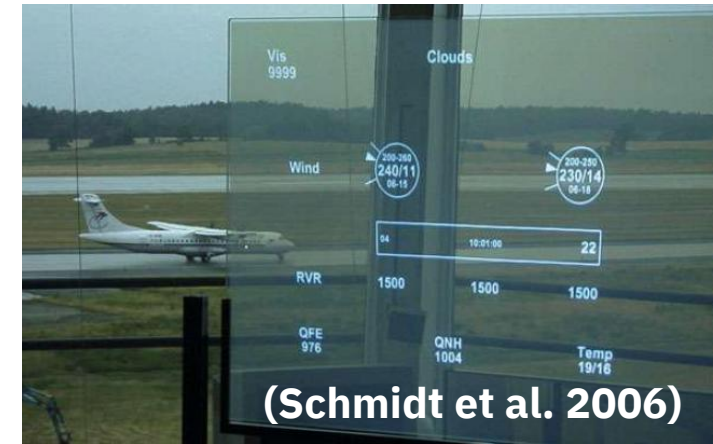
1) https://www.tc.faa.gov/act4/insidethefence/2006/0102_14_tower_tech.htm

2) <https://www.faa.gov/newsroom/faa-commissions-new-air-traffic-control-tower-charlotte-douglas-international-airport>

Remote ATC Center: brief history

First studies: augment the window view with optronic devices (Fürstenau et al., 2004)

Goal: keep controller's focus outside the tower



Then: remove the physical tower and replace it with a remote center (Schaik et al., 2016)

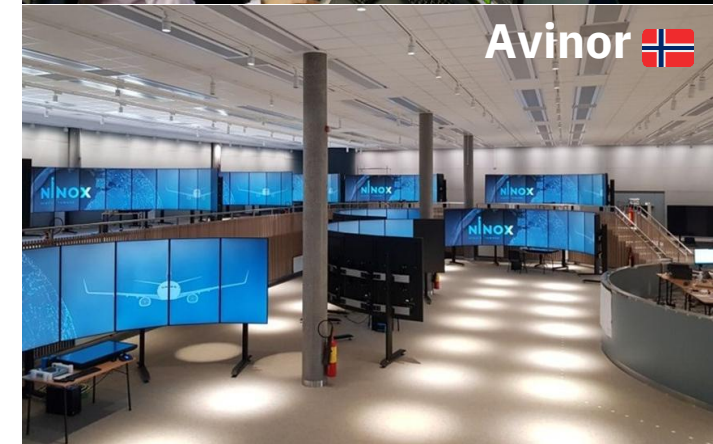
Potentially located hundreds of kilometers away

A screen wall has been substituted for out-the-view windows



Now: one center for multiple airfields (Papenfuss and Friedrich, 2016)

We talk about Multiple Remote Tower Operations (MRTO)



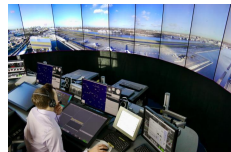
Virtual tower maturity

Description	TRL
Basic principles observed and reported	1
Technology concept and/or application formulated	2
Analytical and experimental critical function and/or characteristic proof-of-concept	3
Component and/or breadboard validation in laboratory environment	4
Component and/or breadboard validation in relevant environment	5
System/subsystem model or prototype demonstration in a relevant environment (ground or space)	6
System prototype demonstration in a space environment	7
Actual system completed and “flight qualified” through test and demonstration (ground or space)	8
Actual system “flight proven” through successful mission operations	9

TRL : Technology Readiness Levels
(Mankins, 1995)

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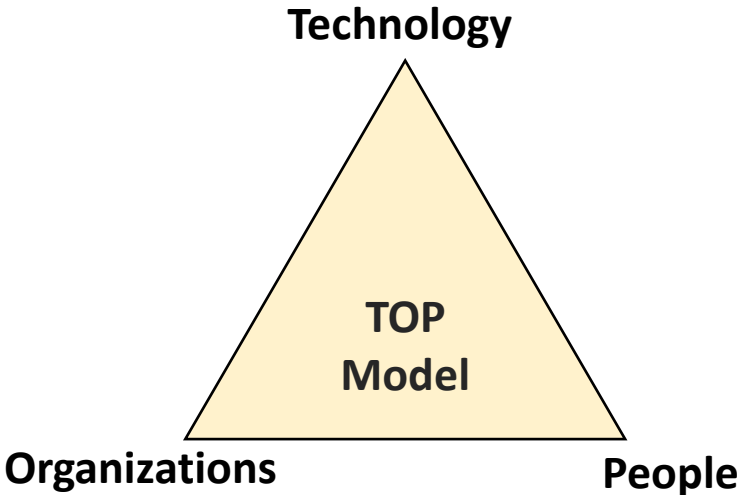
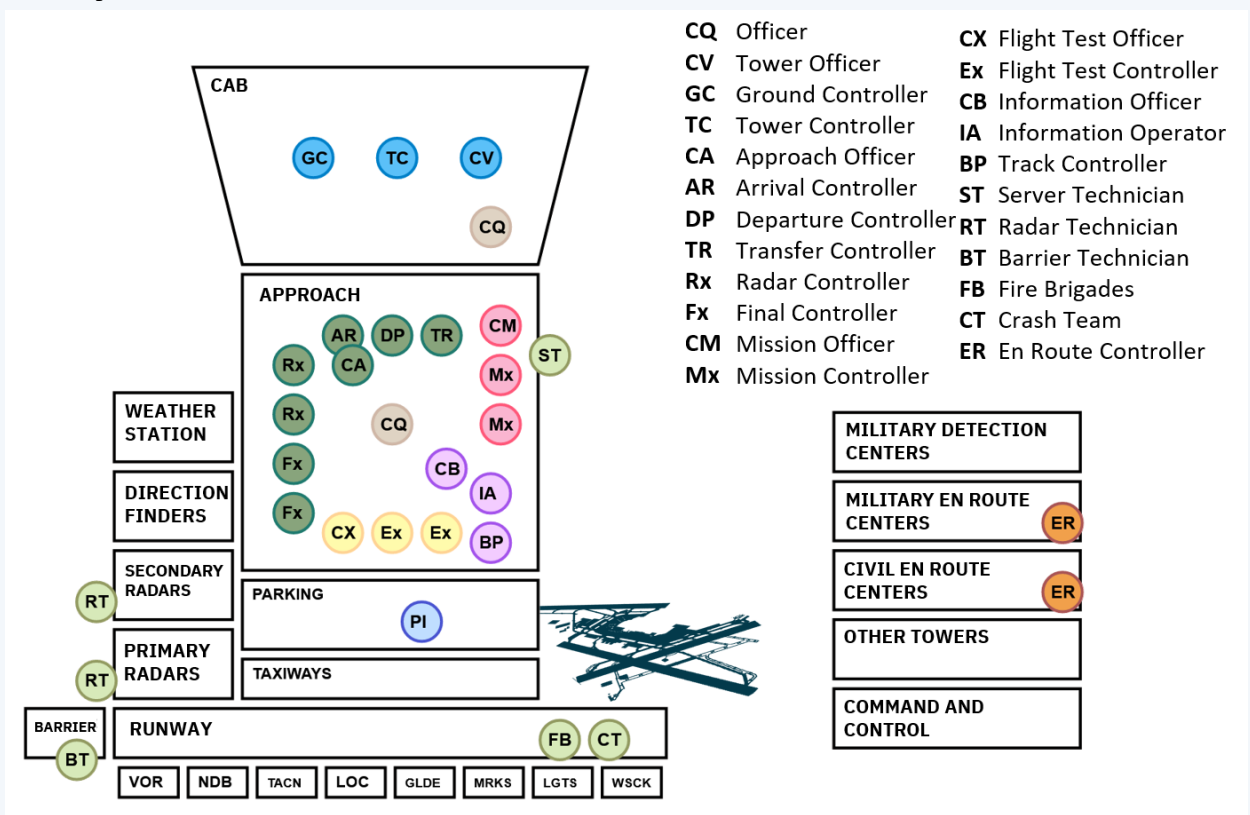


HRL	Description
1	Relevant human capabilities, limitations, and basic human performance issues and risks identified
2	Human-focused concept of operations defined and human performance design principles established
3	Analyses of human operational, environmental, functional, cognitive, and physical needs completed, based on proof of concept
4	Modeling, part-task testing, and trade studies of user interface design concepts completed
5	User evaluation of prototypes in mission-relevant simulations completed to inform design
6	Human-system interfaces fully matured as influenced by human performance analyses, metrics, prototyping, and high-fidelity simulations
7	Human-system interfaces fully tested and verified in operational environment with system hardware and software and representative users
8	Total human-system performance fully tested, validated, and approved in mission operations, using completed system hardware and software and representative users
9	System successfully used in operations across the operational envelope with systematic monitoring of human-system performance

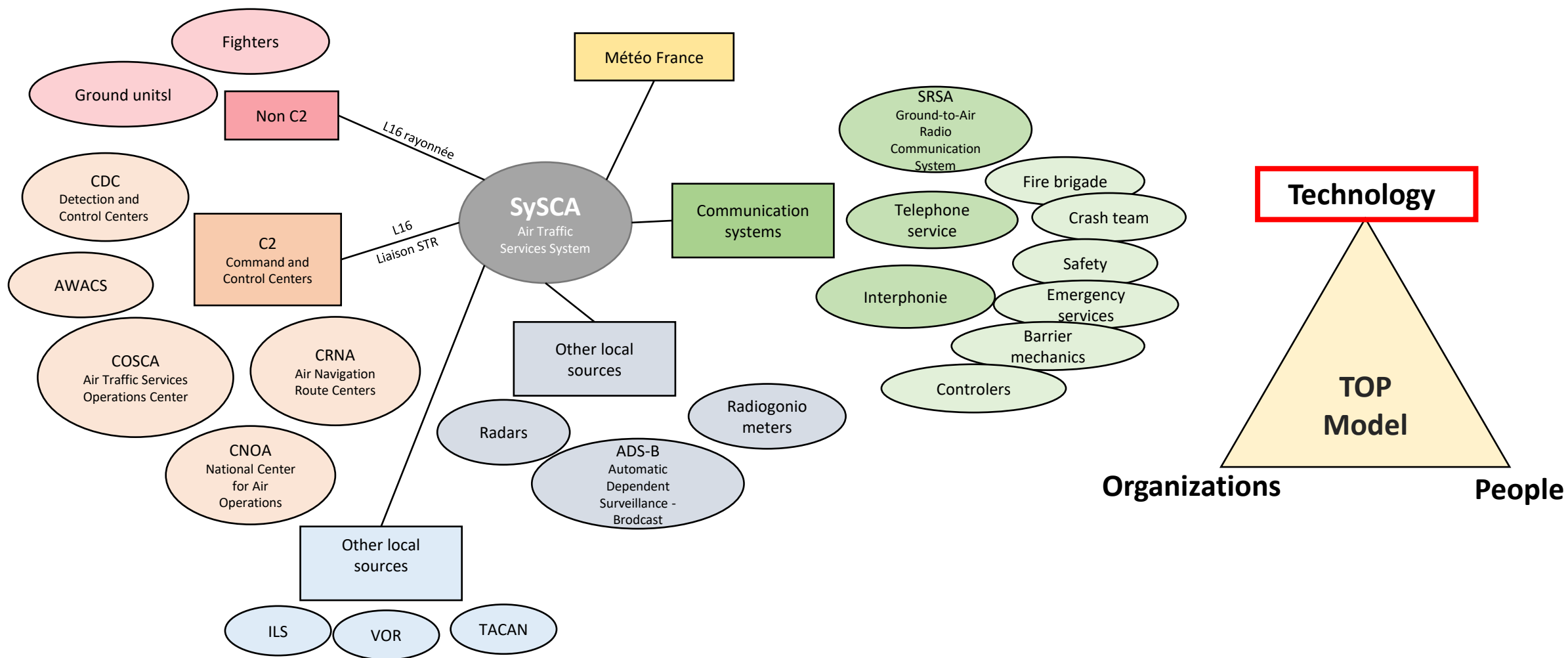
HRL : Human Readiness Levels
(Salazar, 2020)

Air traffic control towers/centers are complex sociotechnical systems

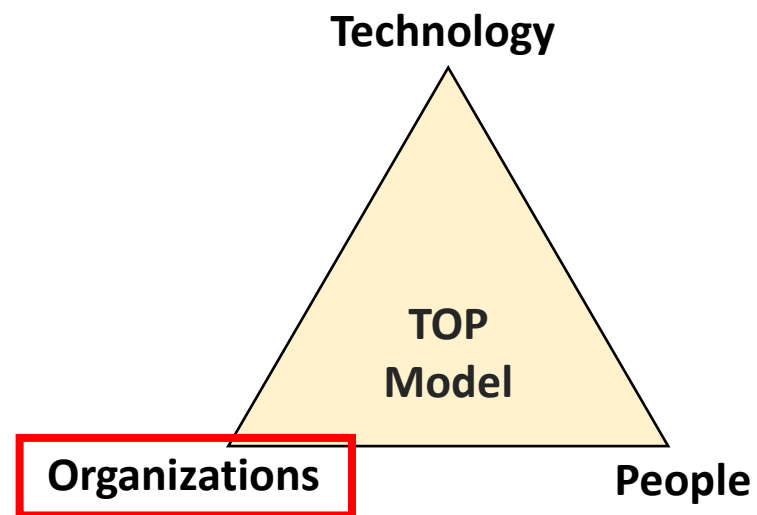
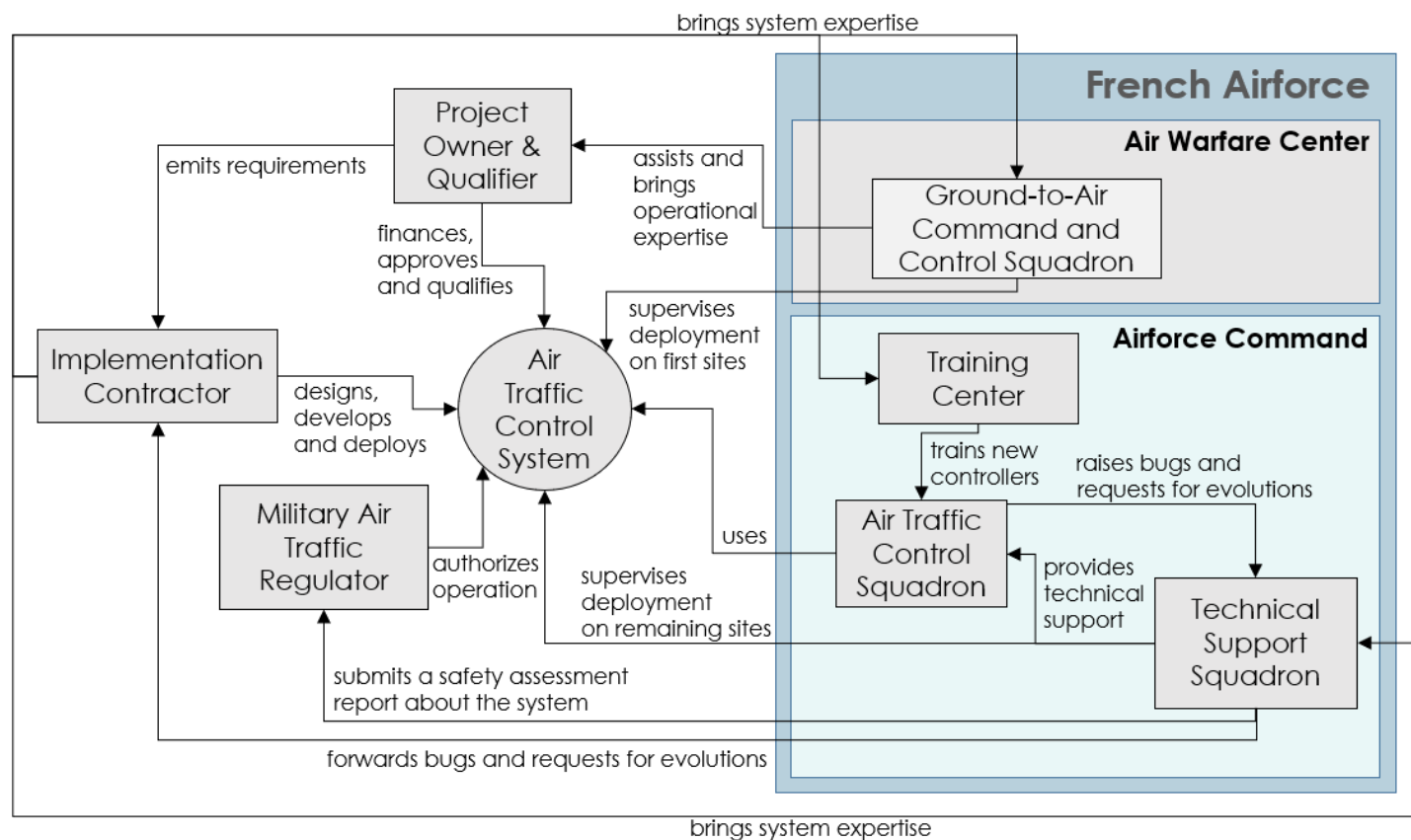
Military tower



Technological complexity of an air traffic control tower



Organizational complexity in control towers



Human complexity in control towers

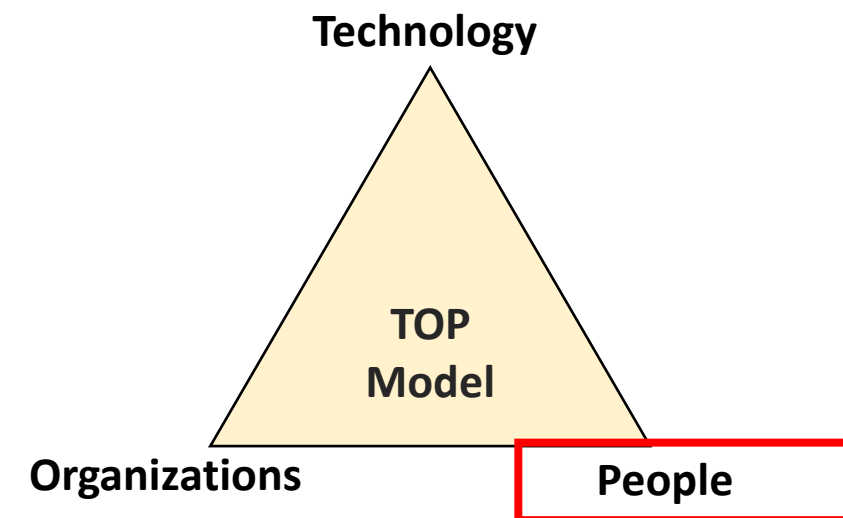
A control tower mobilizes:

Operational people

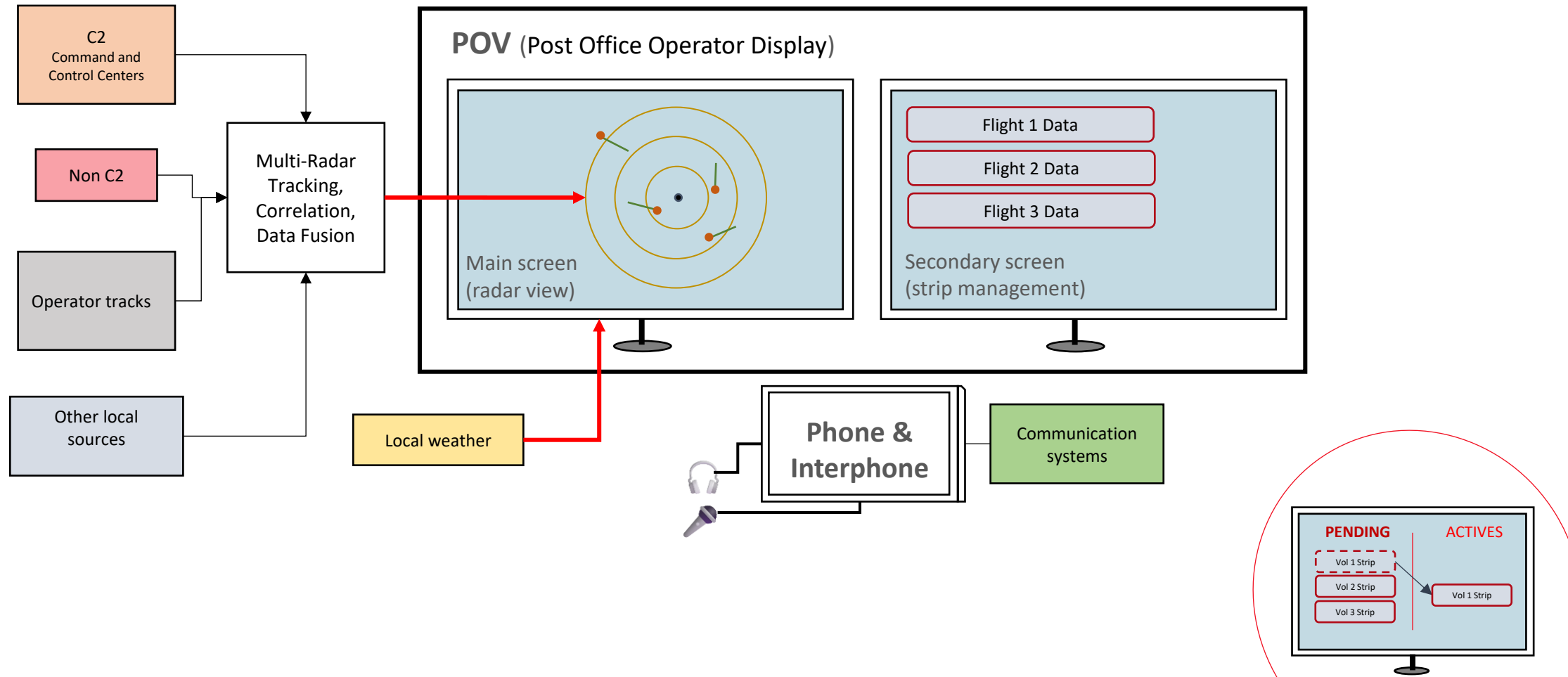
Inspectors, radar technicians, barrier technicians, maintenance personnel, ground personnel, pilots, trainers, weather services, firefighters, emergency services, crash team, safety services

Non-operational people

Designers, developers, testers, engineers, managers, supervisors, trainers, manufacturers, suppliers, qualifiers, regulators



What does an air traffic controller's job involve?

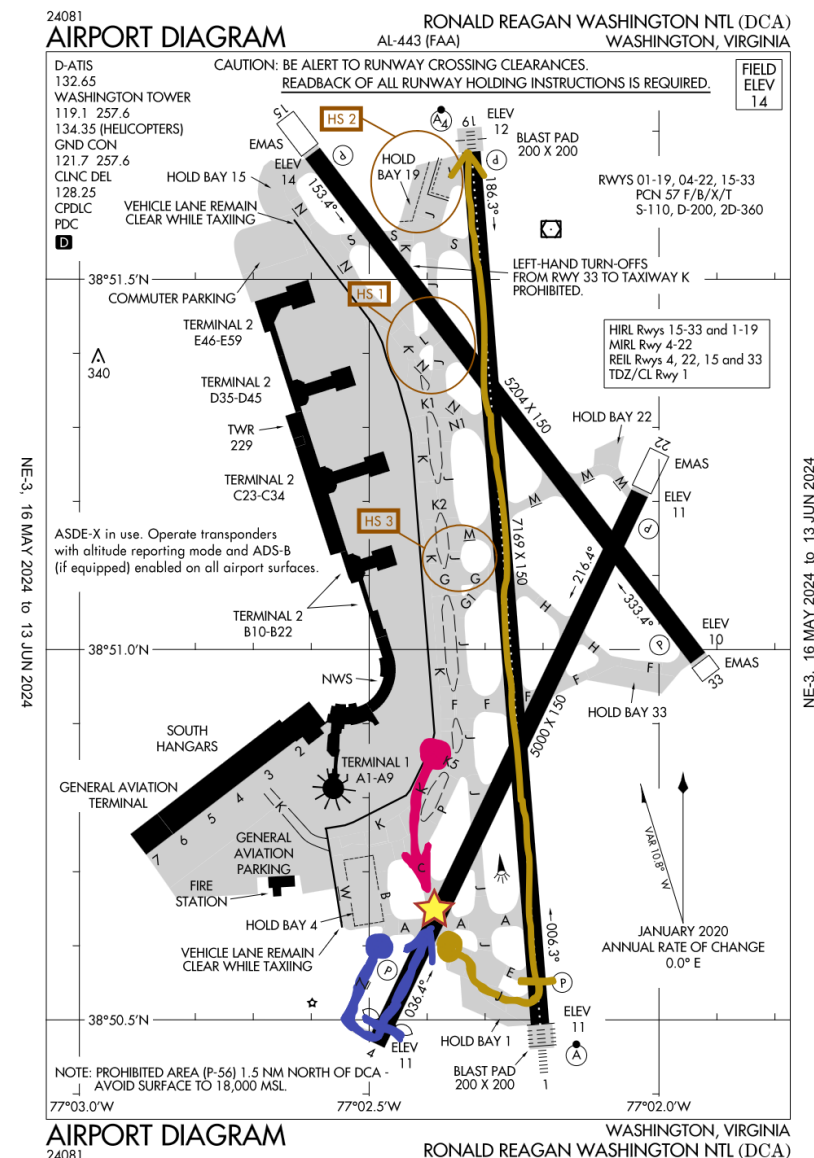


Air traffic control can be a mess

<https://www.youtube.com/watch?v=1LrDIimibNKc>

Date: April 18, 2024. Location: Ronald Reagan Washington National Airport (DCA/KDCA)

A Southwest Airlines Boeing 737 MAX 8 registration N8710M, performing flight SWA2937 from Washington Reagan National Airport (KDCA) to Orlando International Airport (KMCO) was cleared to taxi to the runway 1. On their taxi route the airplane was cleared to cross runway 4. Meanwhile a JetBlue Airways Embraer ERJ-190 registration N323JB, flight JBU1554 from Washington Reagan National Airport (KDCA) to Boston Logan International Airport (KBOS) was waiting for takeoff clearance on runway 4. When Southwest 2937 was approaching runway 4, JetBlue 1554 was cleared for take-off. As the SWA2937 crossed the hold short line the Ground controller instructed them to stop. Meanwhile the Tower controller stopped JetBlue 1554. As it was reported the Southwest stopped about 40 meters/130 feet past the hold short line.

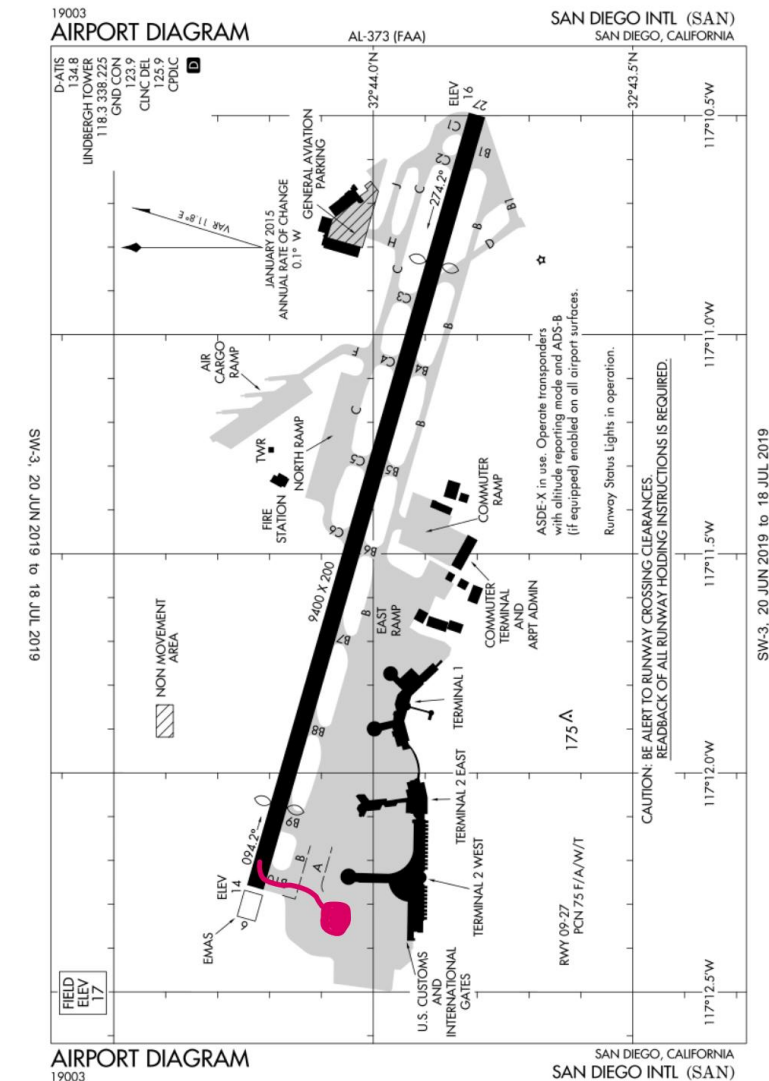


Air traffic control can be a mess

<https://www.youtube.com/watch?v=J3SyhqV5Ns0>

Date: Feb 6, 2024. Location: San Diego, California, USA (KSAN/SAN)

A Japan Airlines Boeing 787-8 Dreamliner (B788), registration JA836J, performing flight JAL65 / JL65 from San Diego International Airport, CA (USA) to Tokyo Narita International Airport (Japan) was taxiing for departure and was instructed to taxi via taxiways B, B8 and hold short of the runway at taxiway B8. But for some reasons the flight crew entered the runway via taxiway B10 without clearance. Meanwhile there was an aircraft on final approach (Delta Air Lines Airbus A220-300 (BCS3), registration N301DU, flight DAL2287 / DL2287) The Tower air traffic controller instructed Delta 2287 to go around. The airplane went around from about 800 feet, distance from touchdown was about 0.54 NM (1 km)

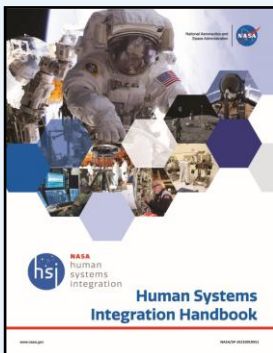


02

Background on HSI approaches

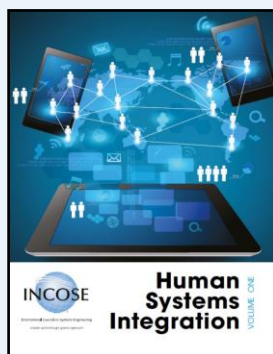
Human Systems Integration (HSI): official definitions

Human Systems Integration (HSI)



(NASA, 2021)

“HSI is the interdisciplinary integration of humans as part of a system to ensure that the human components, software and hardware cooperate, are coordinated and communicate effectively to perform a specific function or mission.”



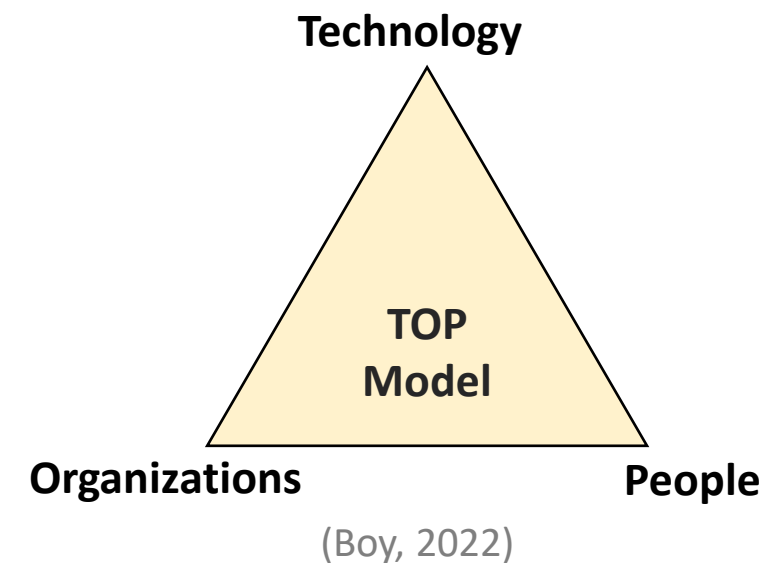
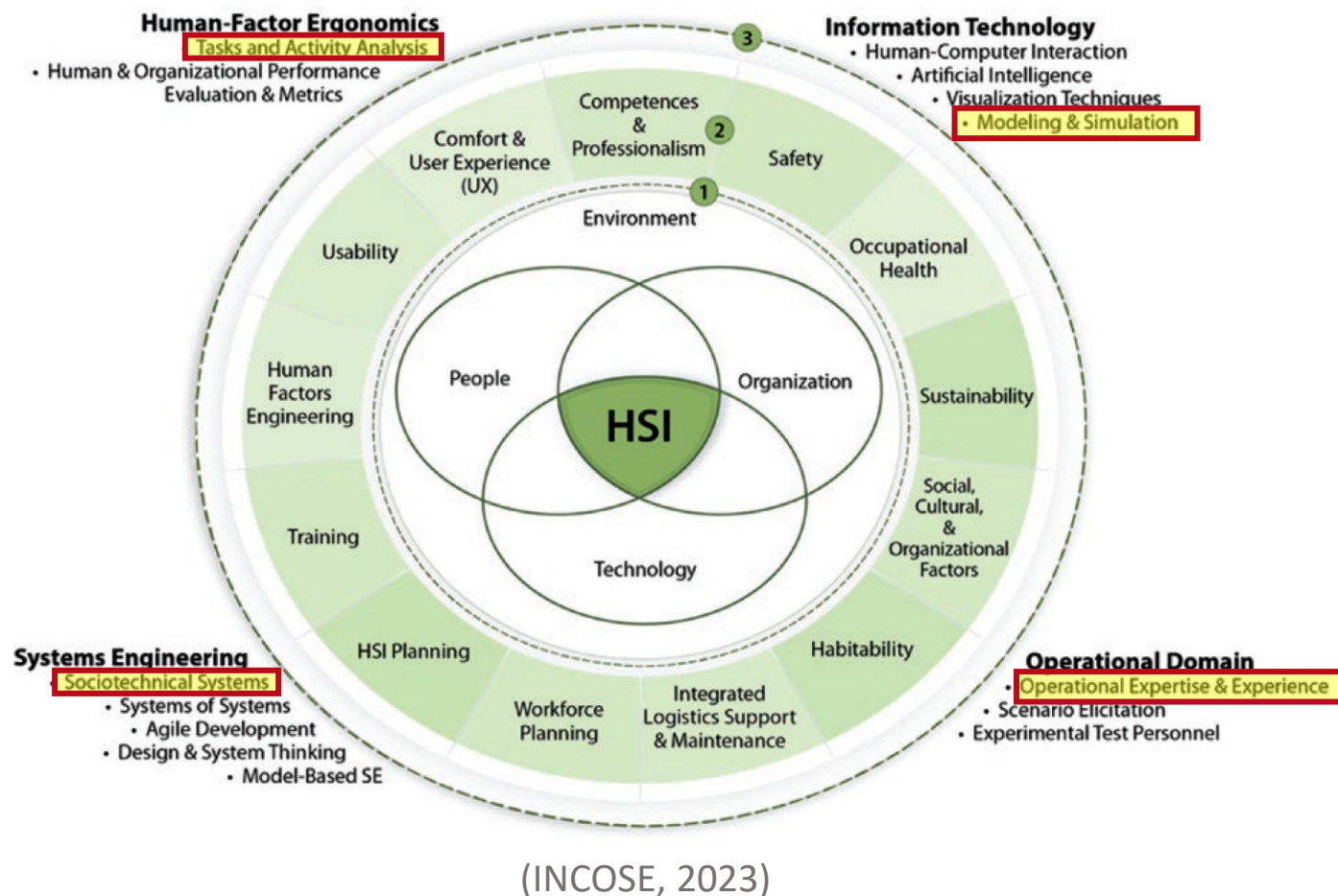
(INCOSE, 2023)

“The objective of the HSI is to recognize the fundamental role of people in complex systems by taking them into consideration and integrating them throughout the entire life cycle of the system.”

“The HSI encourages an understanding of the human roles, their responsibilities, and the limitations and constraints that can influence the design of the system.”

(Handley, 2021)

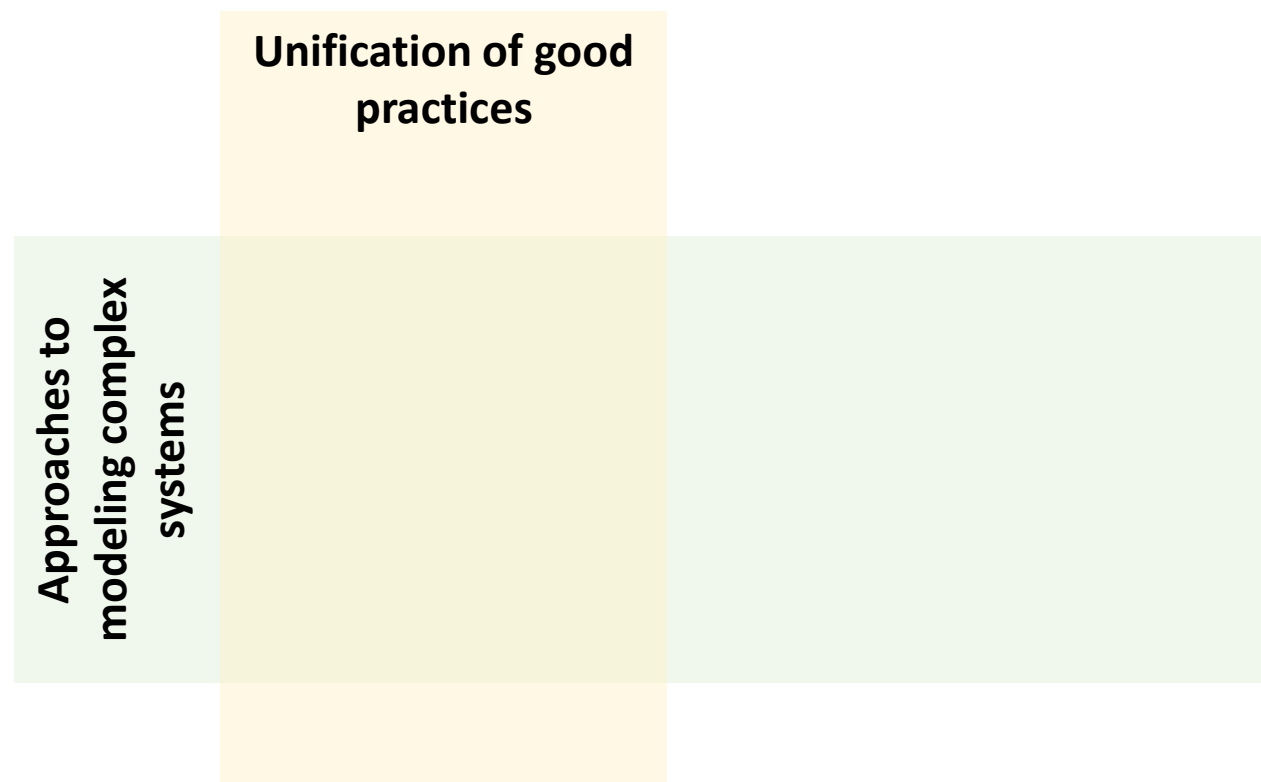
HSI, where several disciplines meet



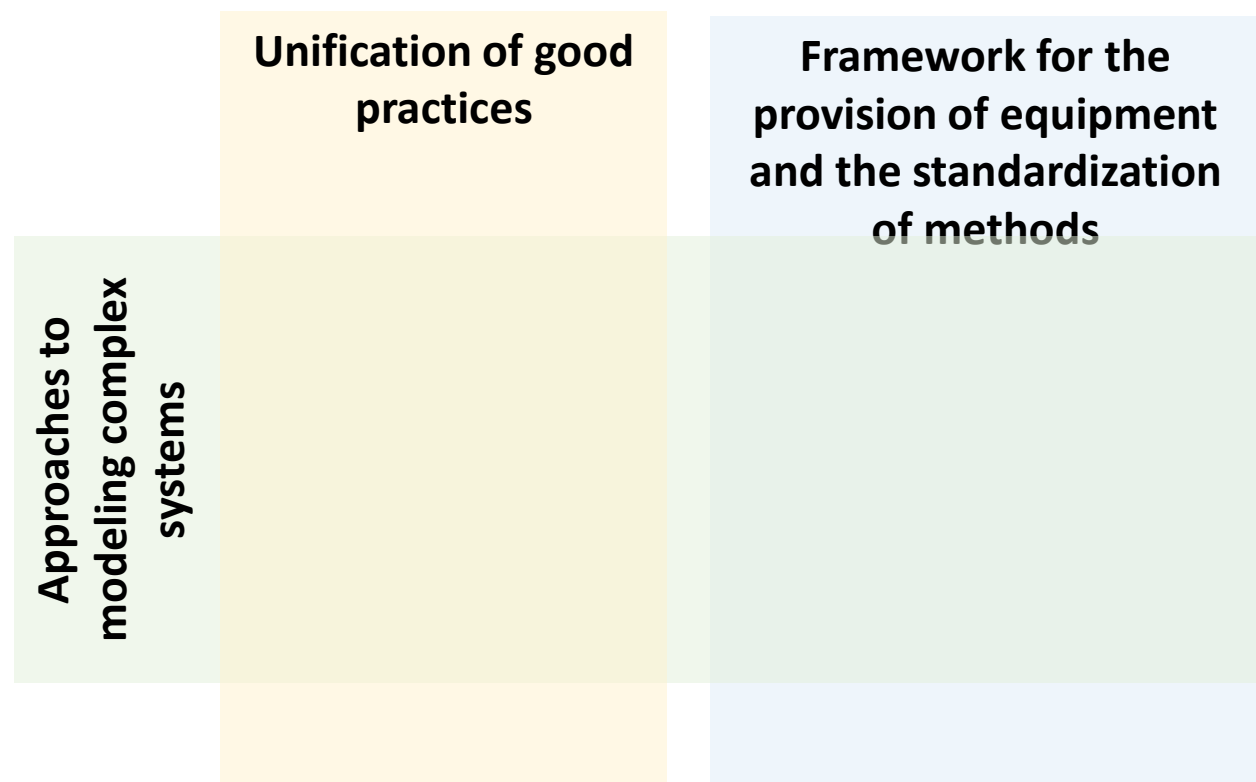
HSI, where several disciplines meet

**Unification of good
practices**

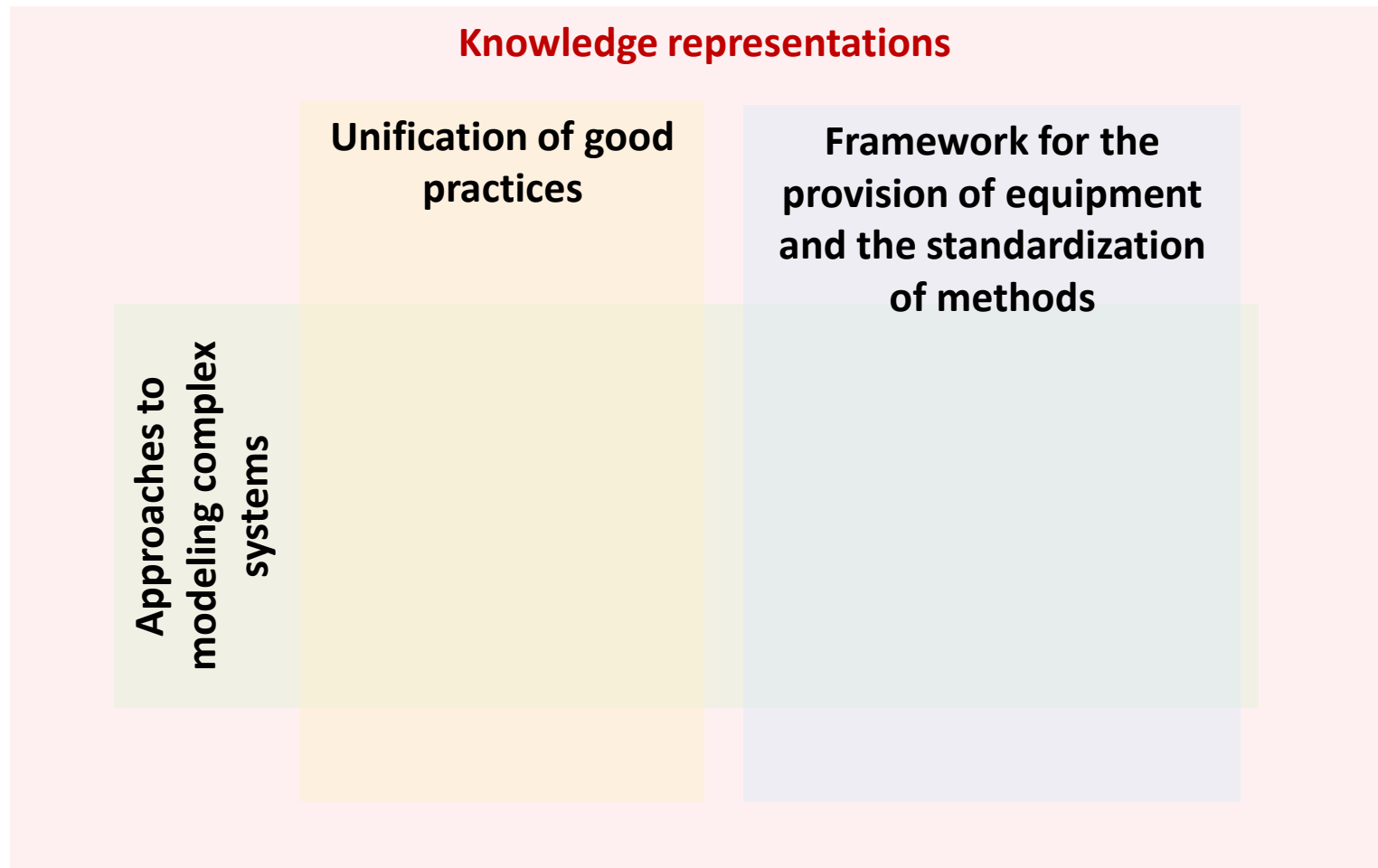
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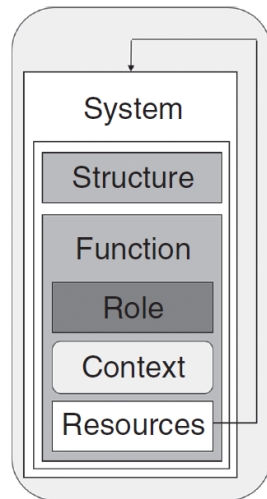
HSI, where several disciplines meet



The socio-technical system in HSI

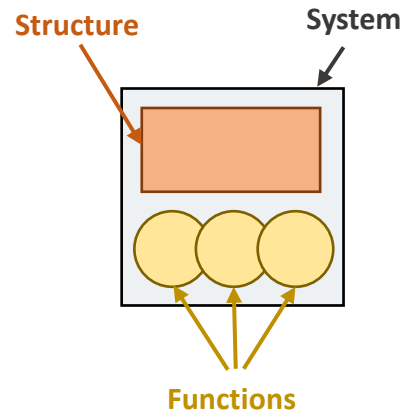
"A system is an arrangement of parts or elements that together exhibit behaviour or meaning that the individual constituents do not."

(INCOSE, 2023)

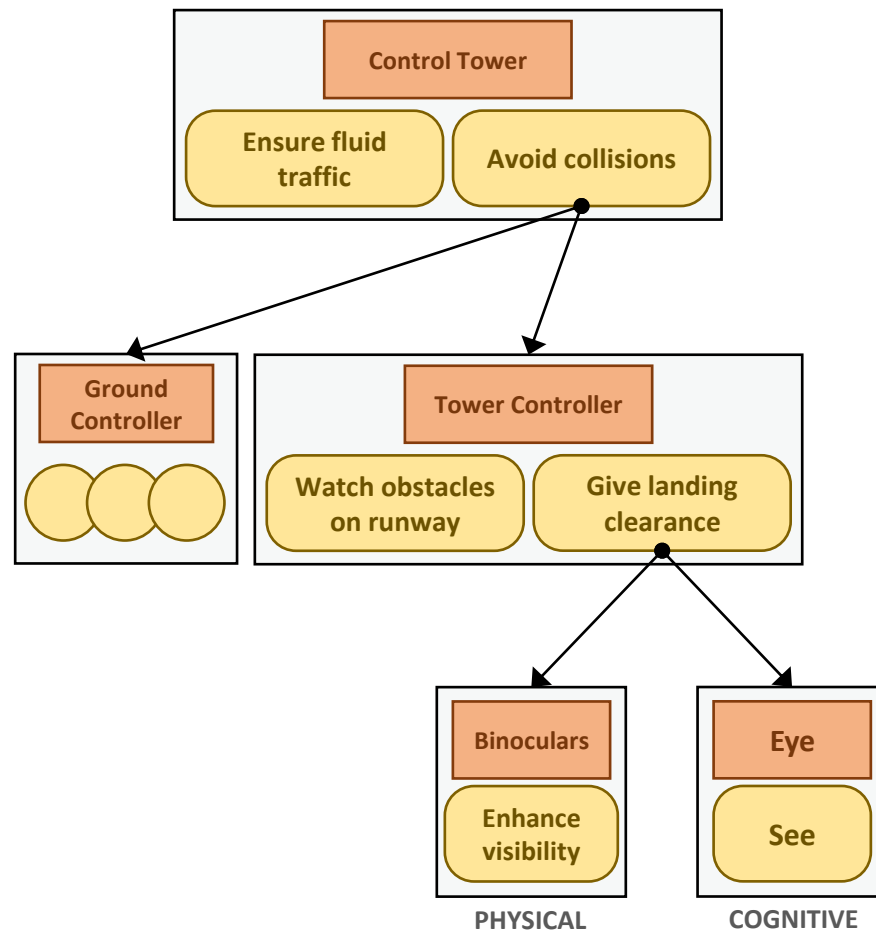
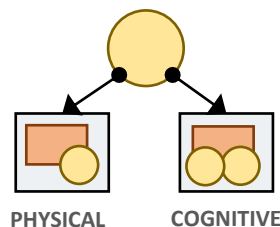


(Boy, 2022)

A system has a structure and functions

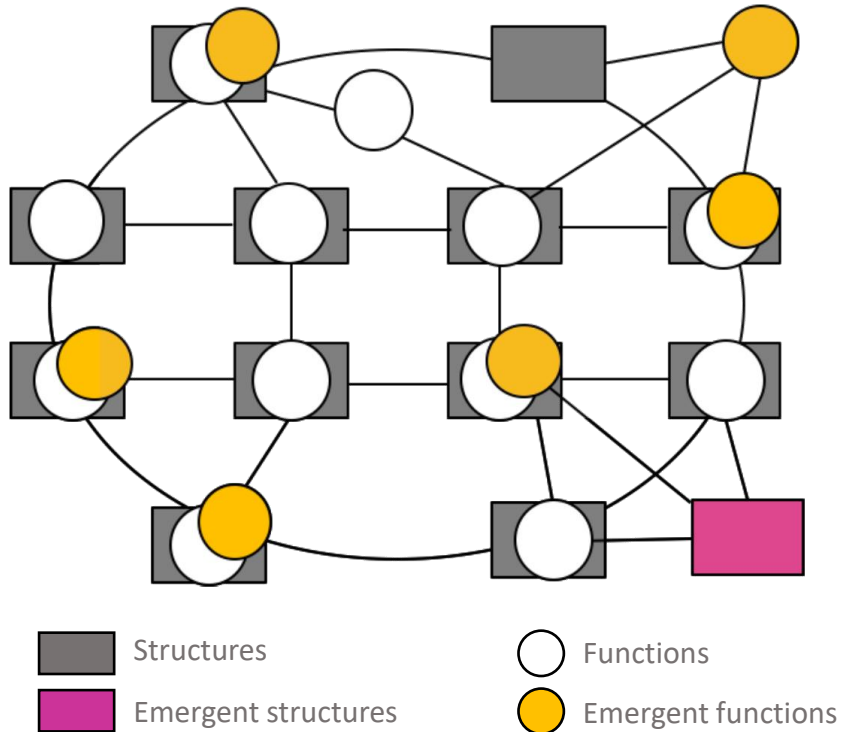


A function has physical or cognitive resources, which are themselves systems



The concept of emergence in HSI

(Boy, 2022)



The emergence paradox

Emerging properties only manifest once the system is active

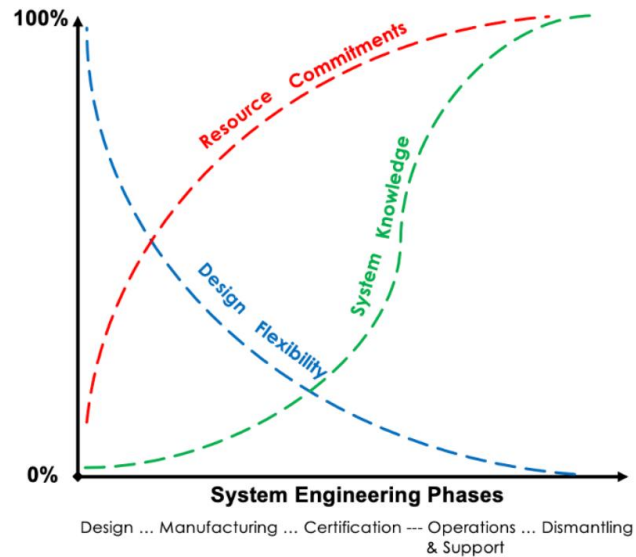
The system can only be observed in operation once integrated

Problem: too often, integration happens once design ends

So, even if emerging properties are useful for design, they too often discovered after design

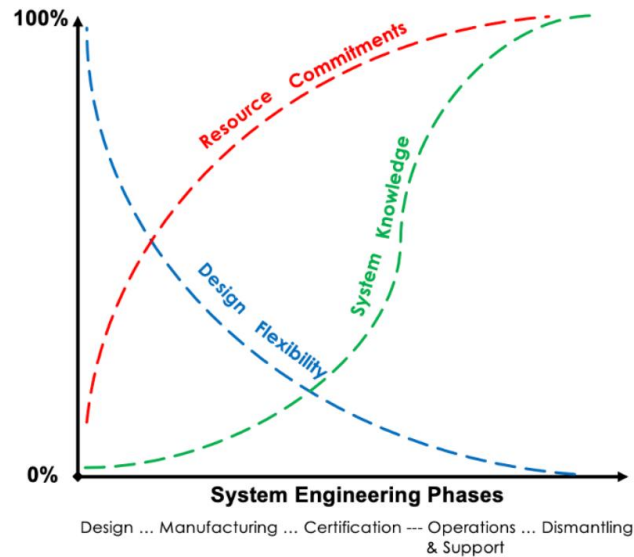
Technology-centered and human-centered approaches

Technology-centered engineering

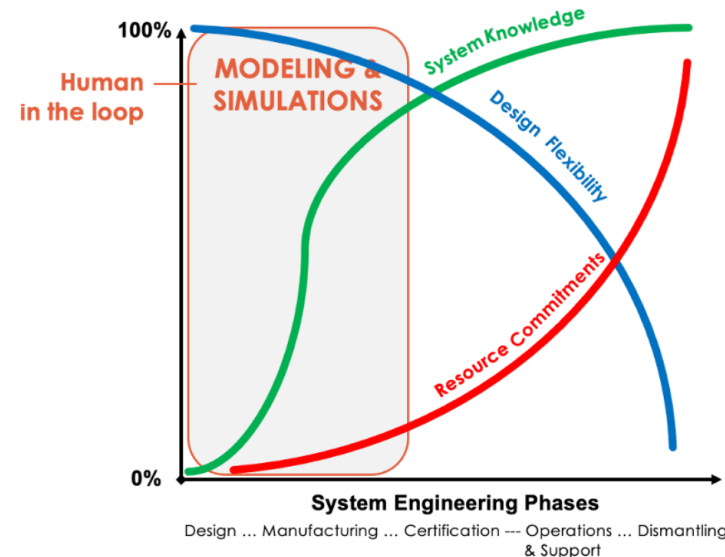


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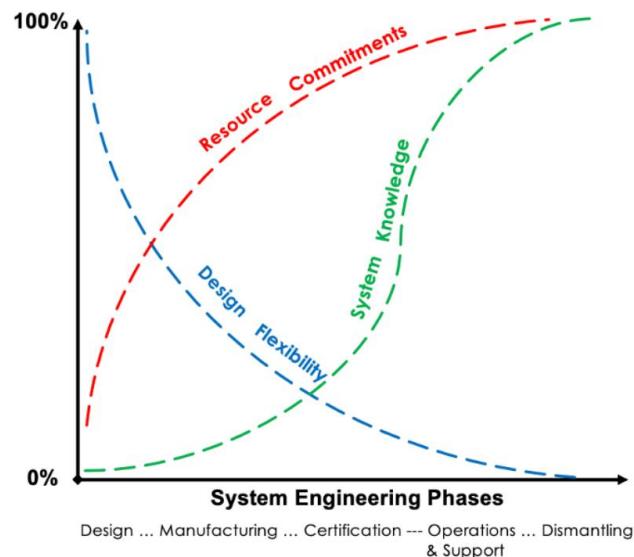


Human-centered design

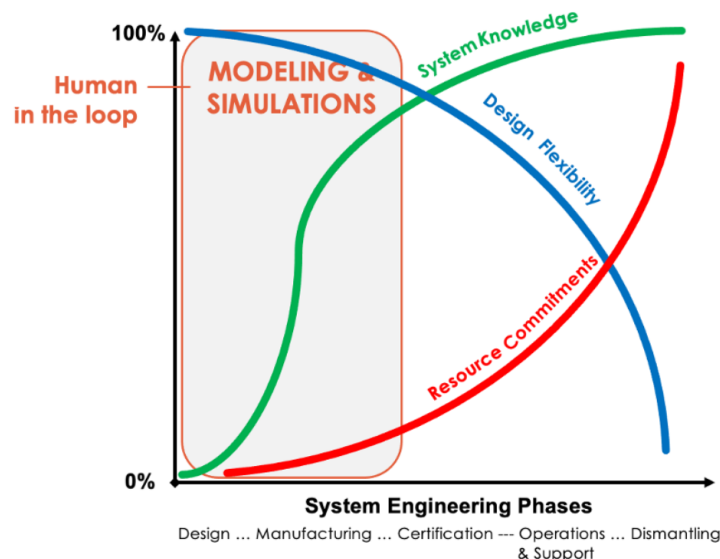


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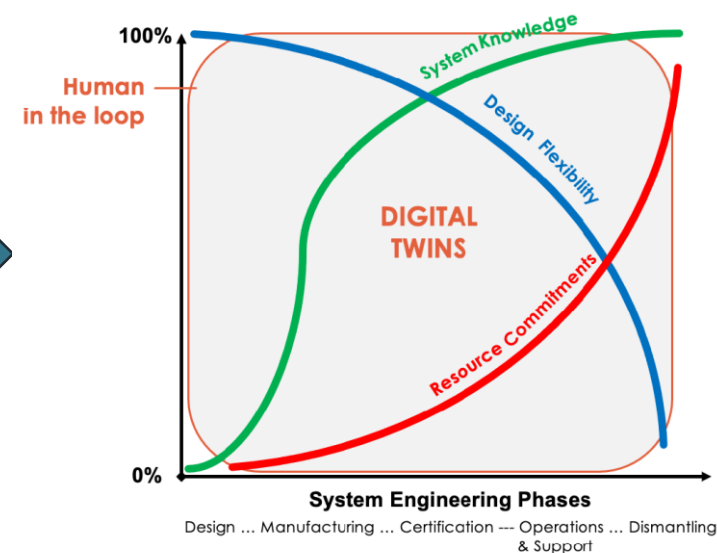
Technology-centered engineering



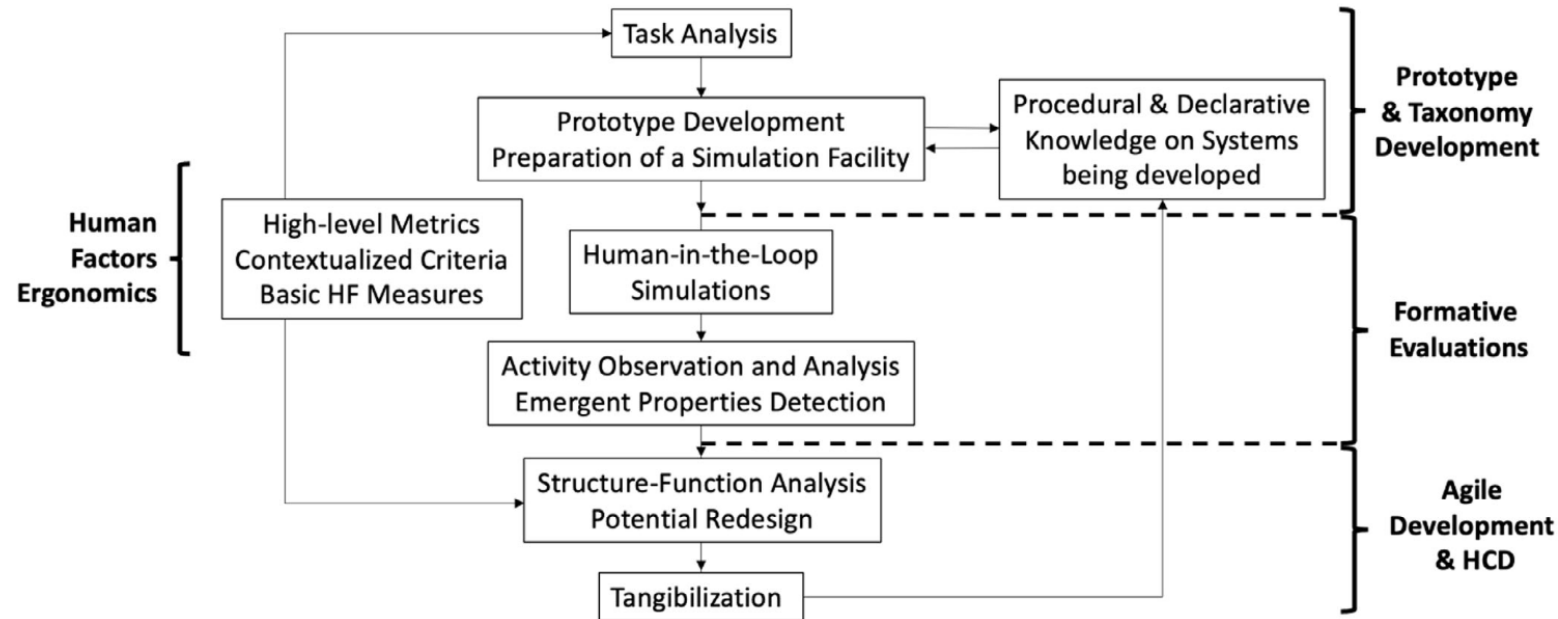
Human-centered design



Human-centered design with digital twins



Activity simulation methodology: PRODEC



Approaches used

Scenario-Based Design (SBD)

“The use of a future system is described in concrete terms, and this early in the design process.”
(Rosson & Carroll, 2002)

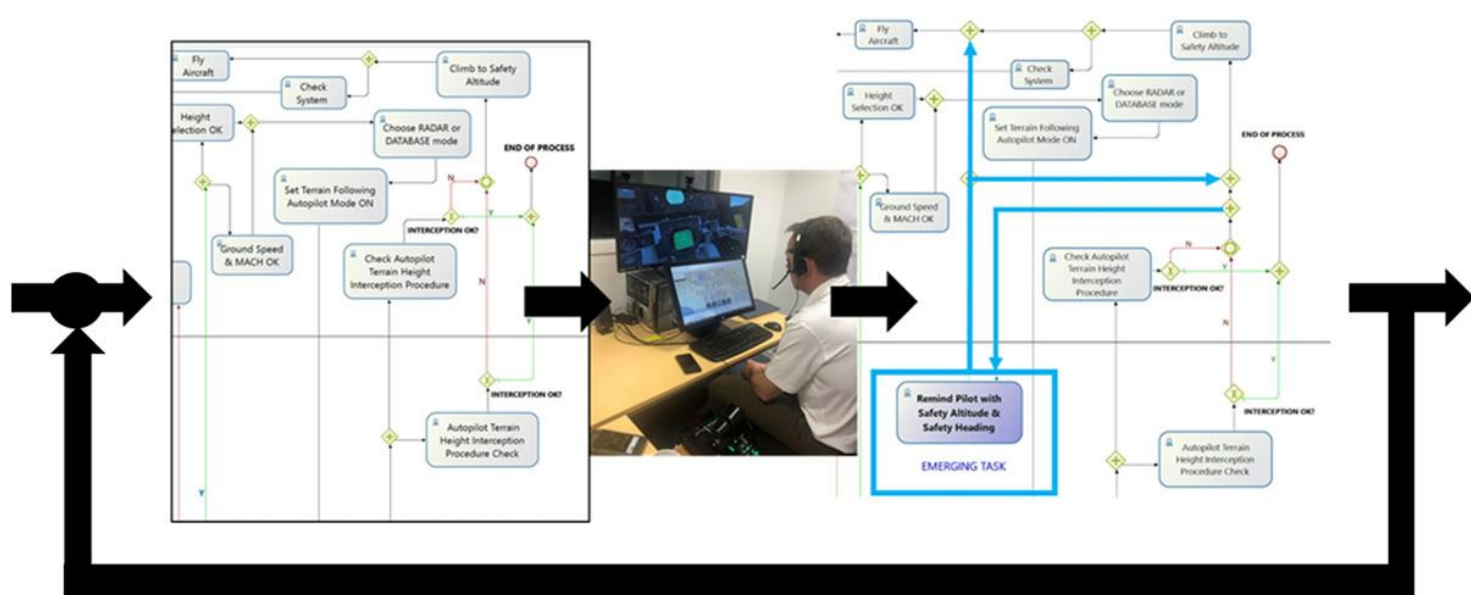
Human-In-The-Loop Simulations (HITLS)

“To gain a better understanding of human behavior in complex situations.”
(Rothrock & Narayanan, 2011)

Example 1: PRODEC and human-machine teaming for the FCAS*



Experimental setup



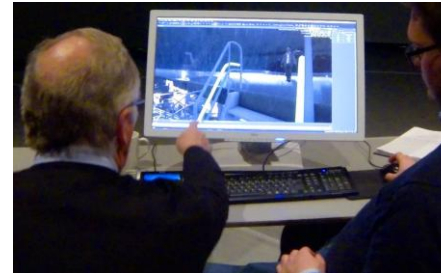
Detection of emerging functions

*FCAS = Future Combat Air System

Example 2: CryEngine for heliport design

Heliports design for the Norway navy(Hjelseth, 2015)

Use of the CryEngine engine for the positioning of light beacons around an offshore heliport



Value of *human-in-the-loop virtual simulations*

- Shared language among designers and operators (**Madni, 2015**)
- Structures and functions can be designed in real time
- No heavy physical resources investment
- Encourage disruptive thinking departing from existing systems schemas
- Increasingly accessible use, even for non-developers
- Playing and replaying scenarios in different contexts
- ⚠ You have to weigh up what is achievable and what is not in real life
- ⚠ Tangibility of the system

Some projects of interest relating to digital simulation



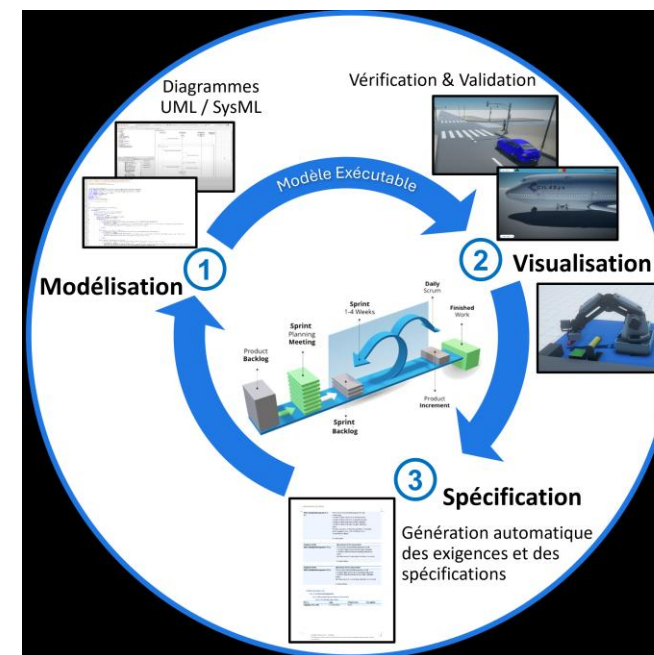
Design of the Falcon 7x using CATIA

© airjet-world.com



Wargaming at DGA CATOD

© defnat.com



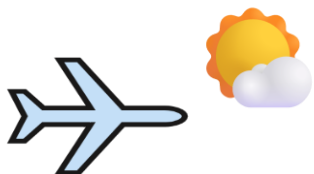
Sim4Sys

© cil4sys.com

3

Context and remote towers

What happens when a plane wants to land on a runway?

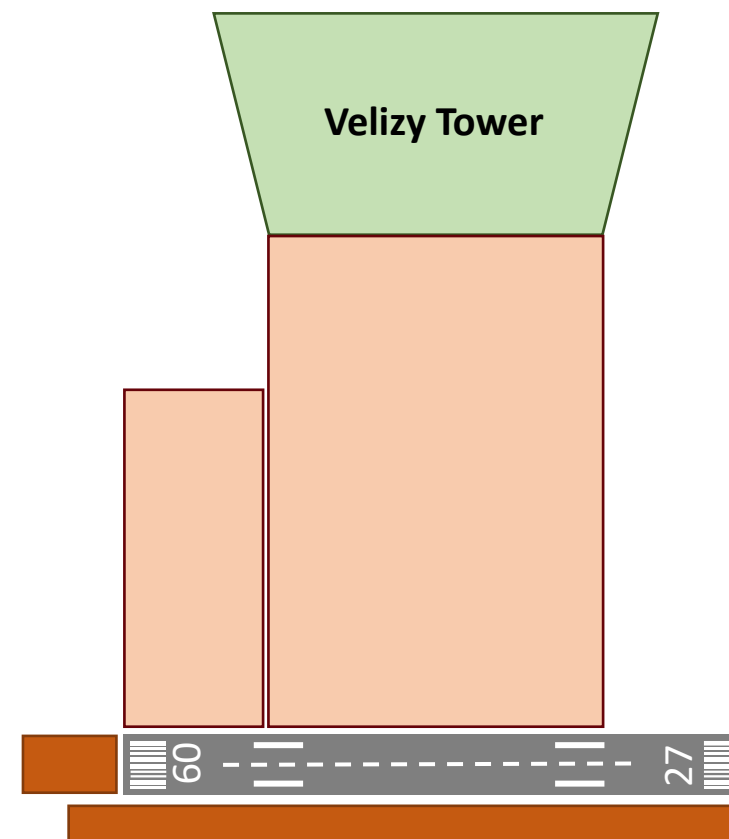


Rafale Airfighter



Velizy Air Base Runway

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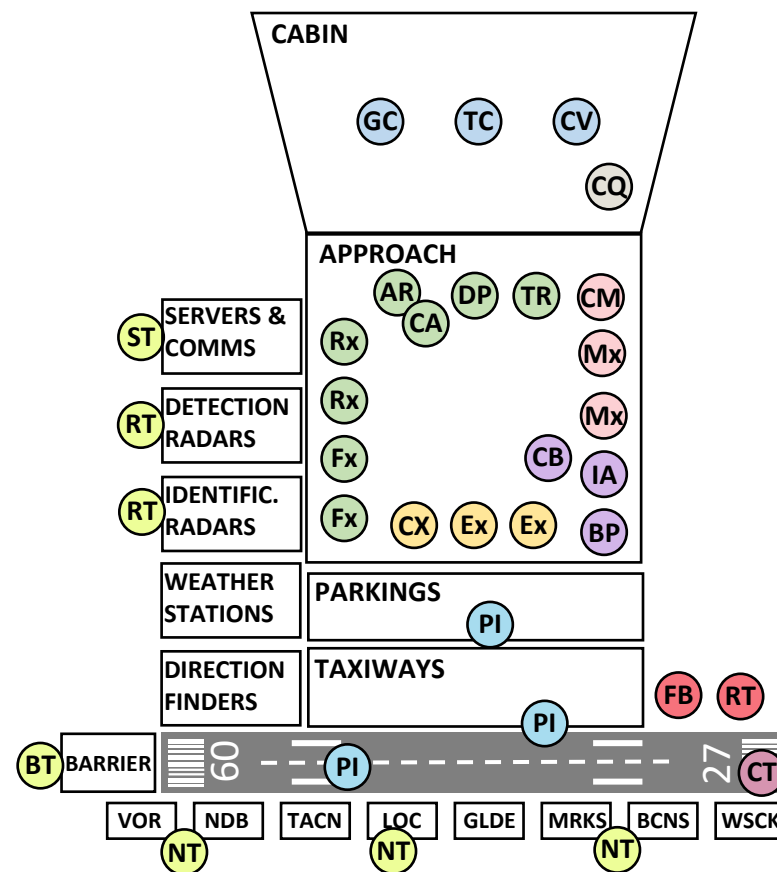


What happens when a plane wants to land on a runway?



*"Air Traffic Control systems
are sociotechnical systems."*

(Foster et al., 2021)



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- Fx Final Controller
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- TC Tower Controller
- GC Ground Controller
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- CA Approach Officer
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- Mx Mission Controller
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- PI Pilots
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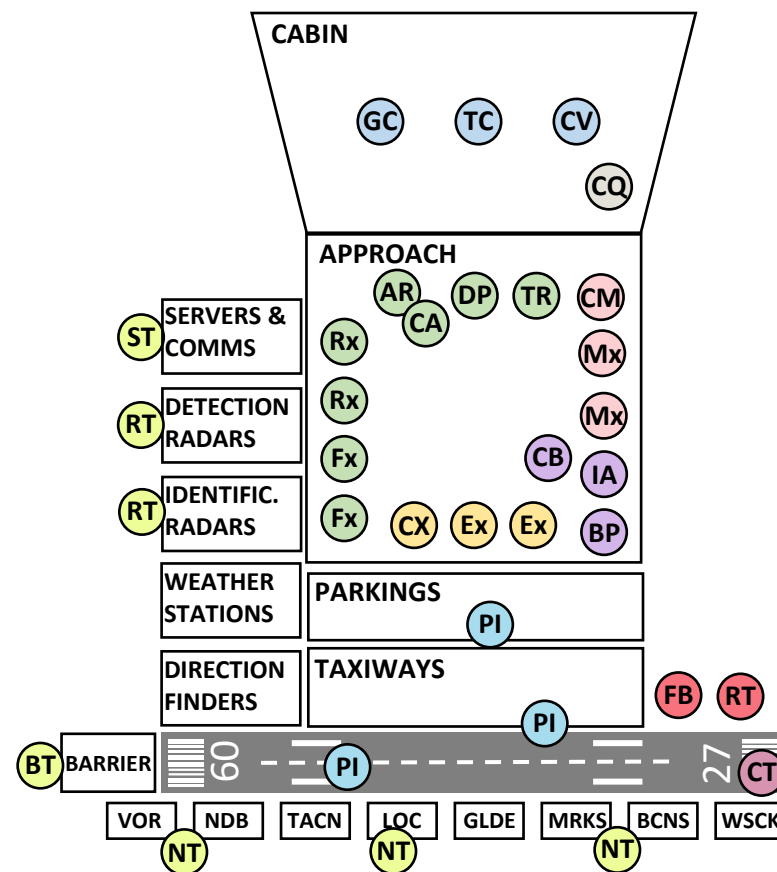
"Air Traffic Control systems are sociotechnical systems."

(Foster et al., 2021)

Sociotechnical System (STS)

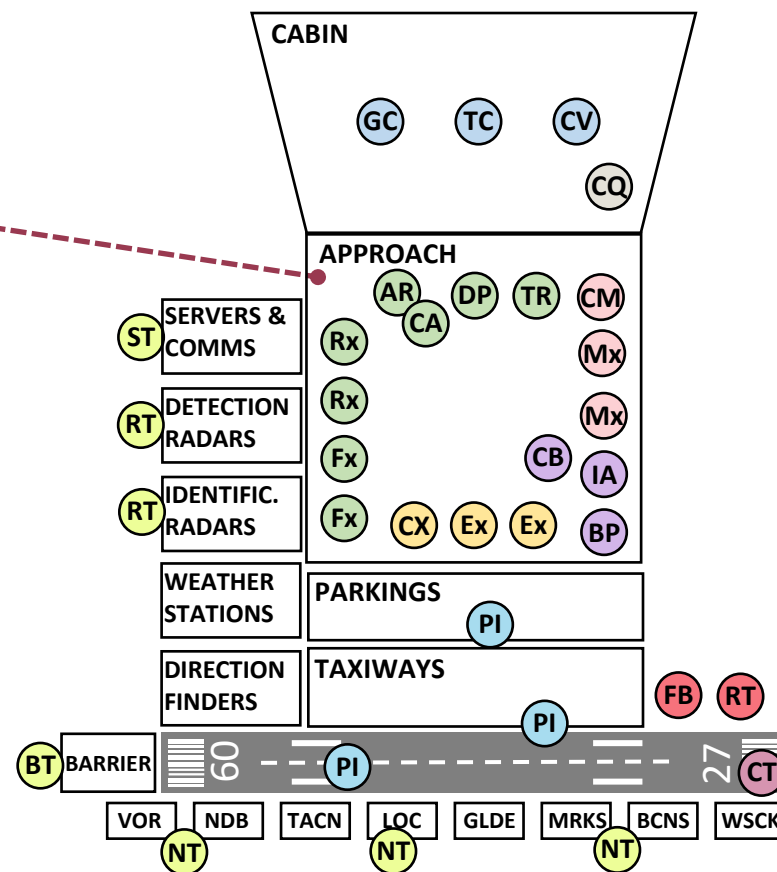
"A system that includes a combination of technical and human or natural elements."

(SEBoK, 2024)



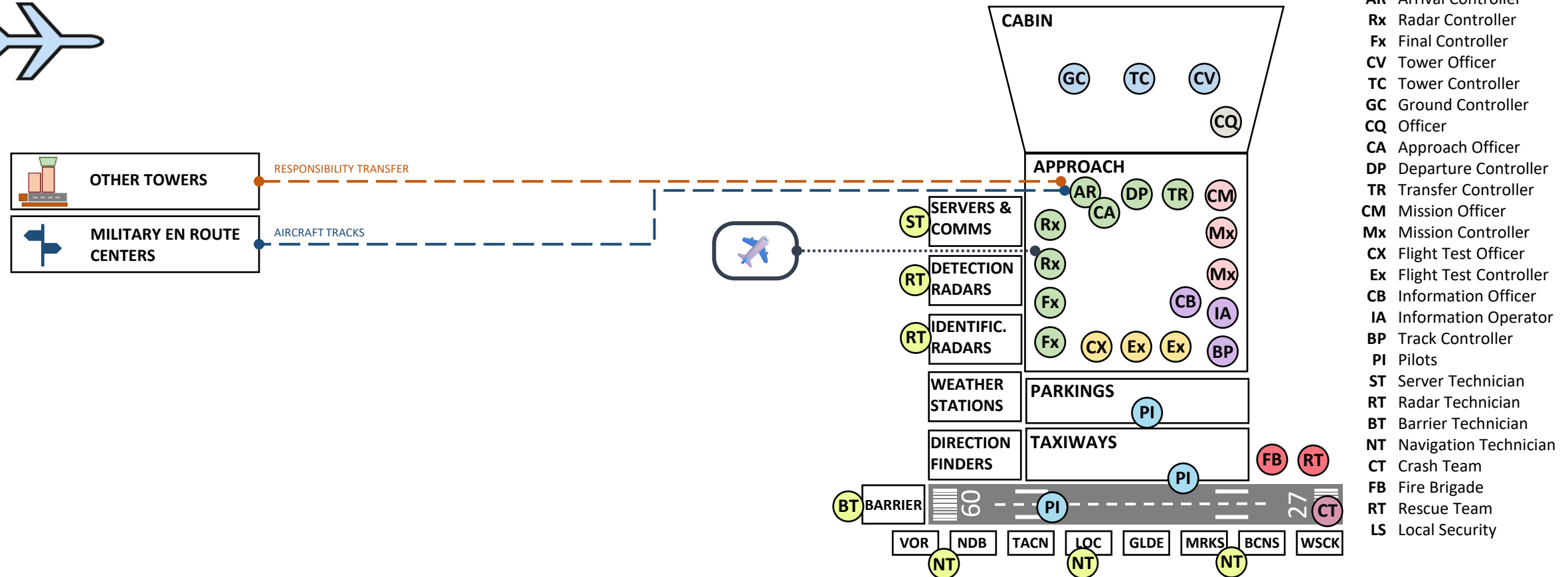
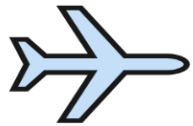
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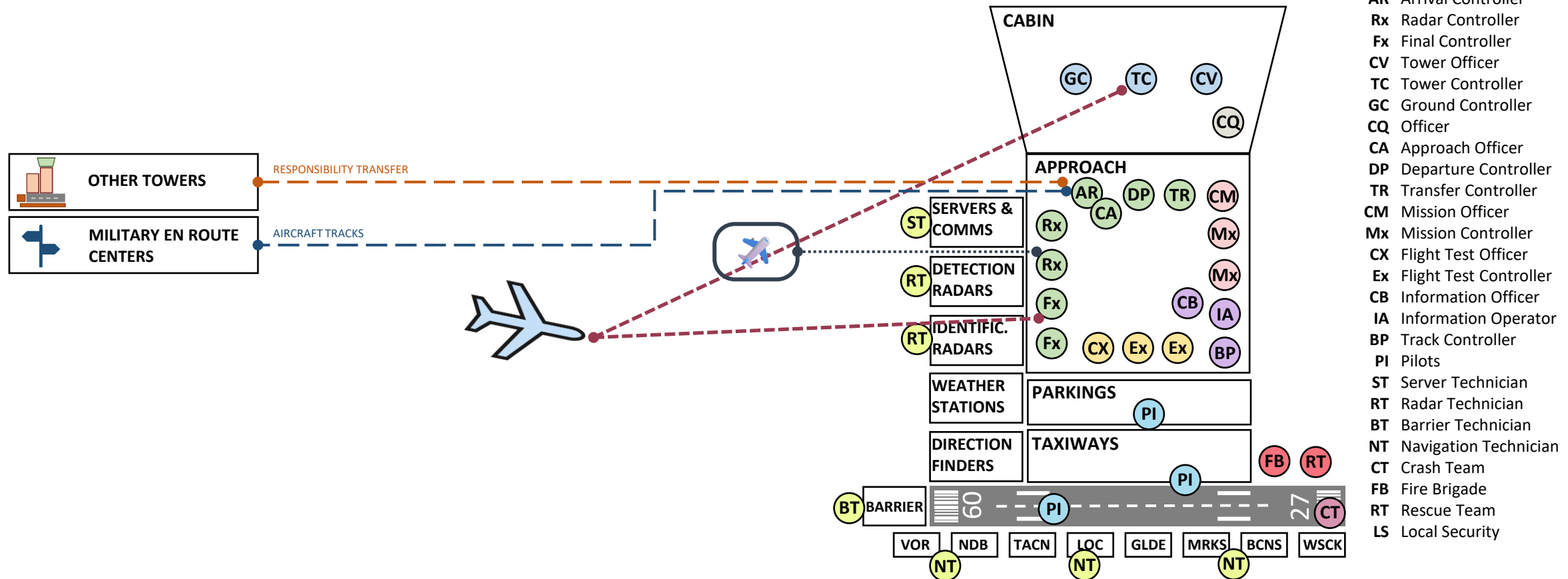


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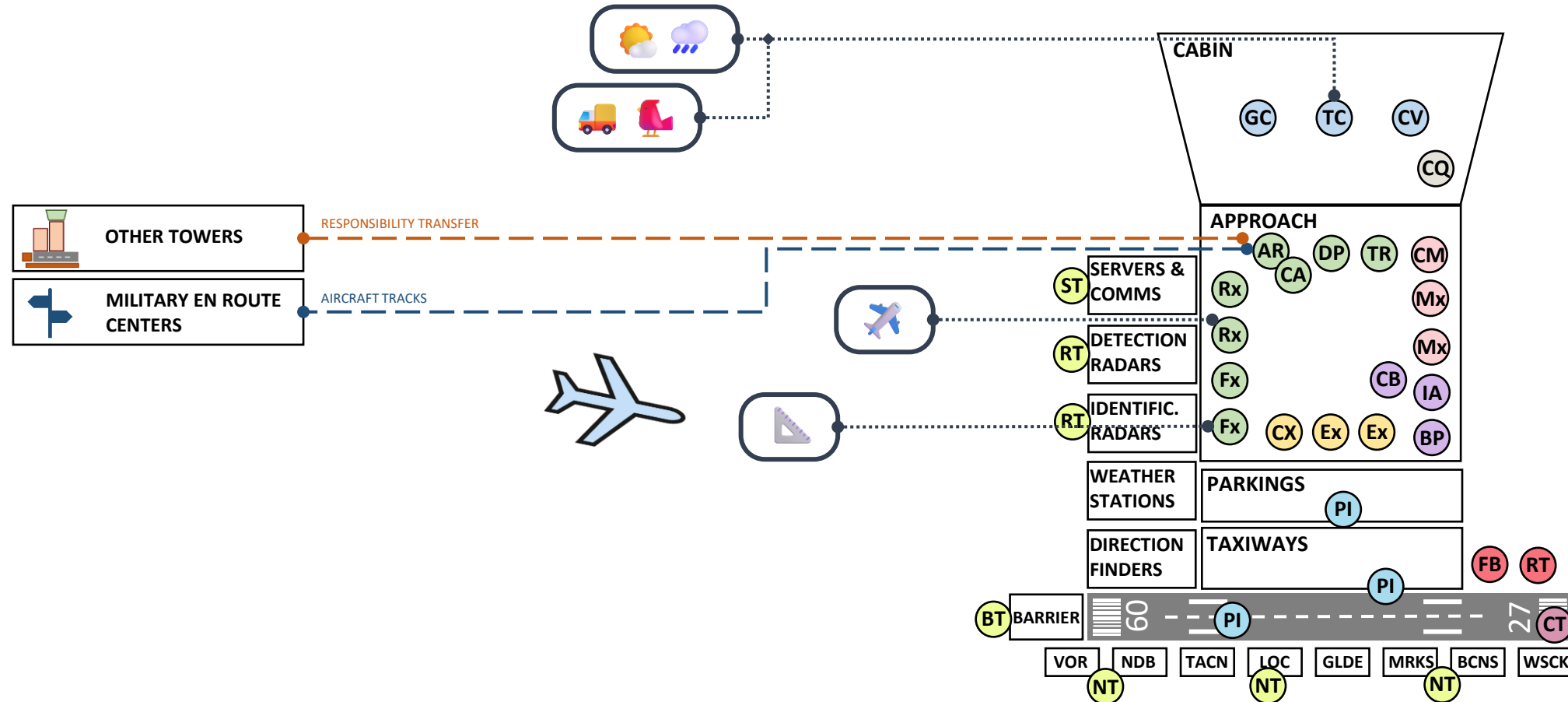
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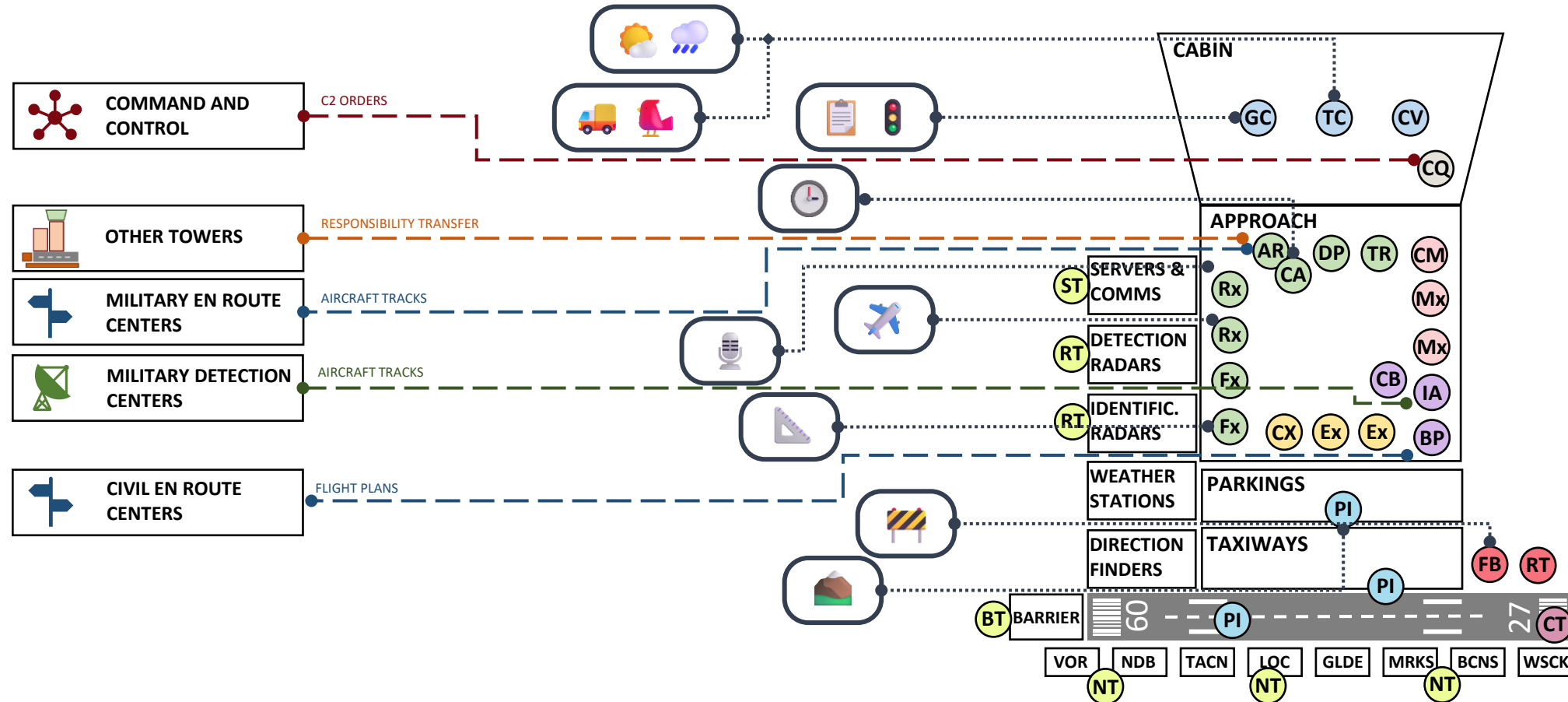


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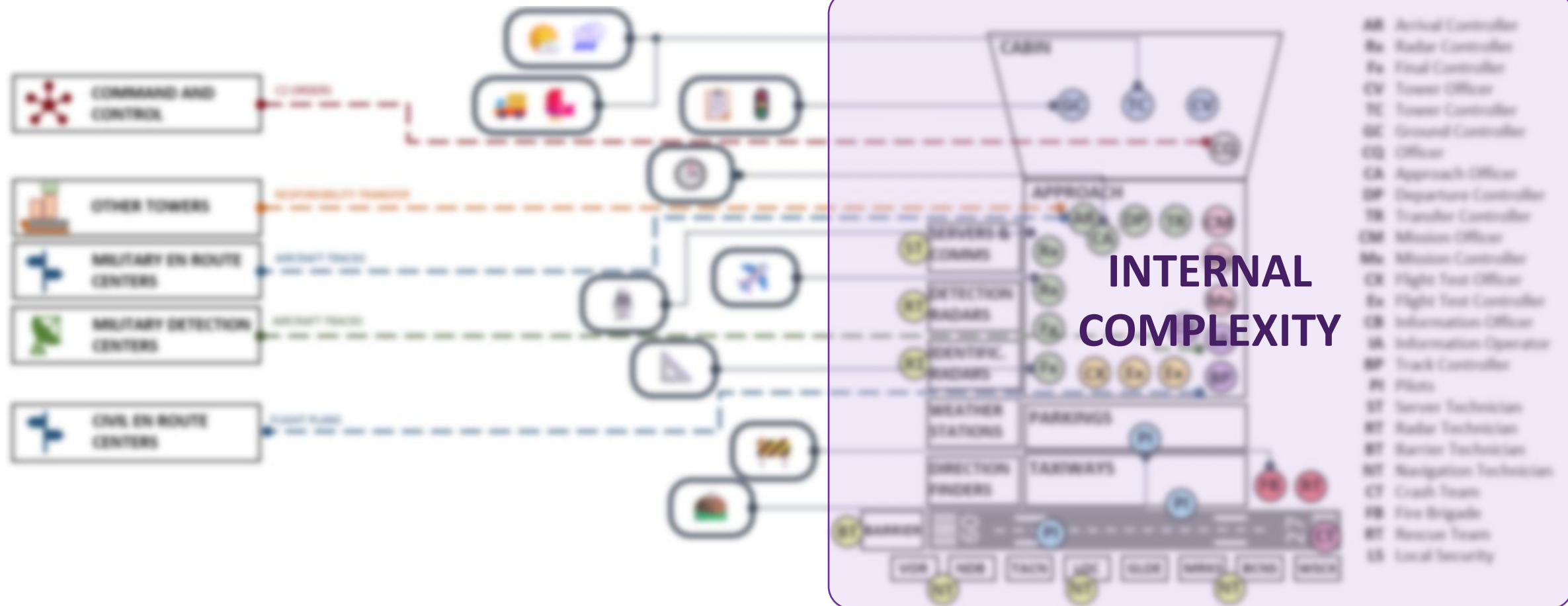
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- LS Local Security

Complexity of air traffic control operations



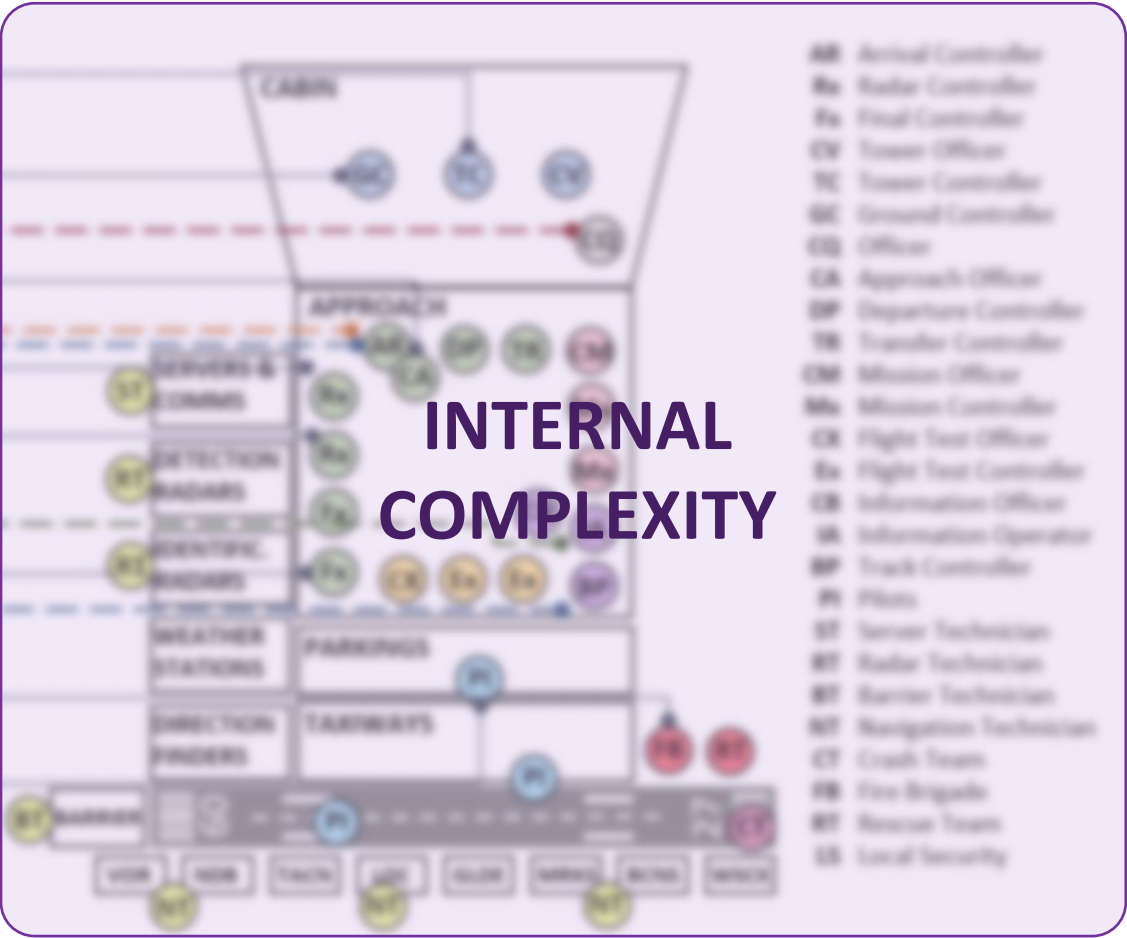
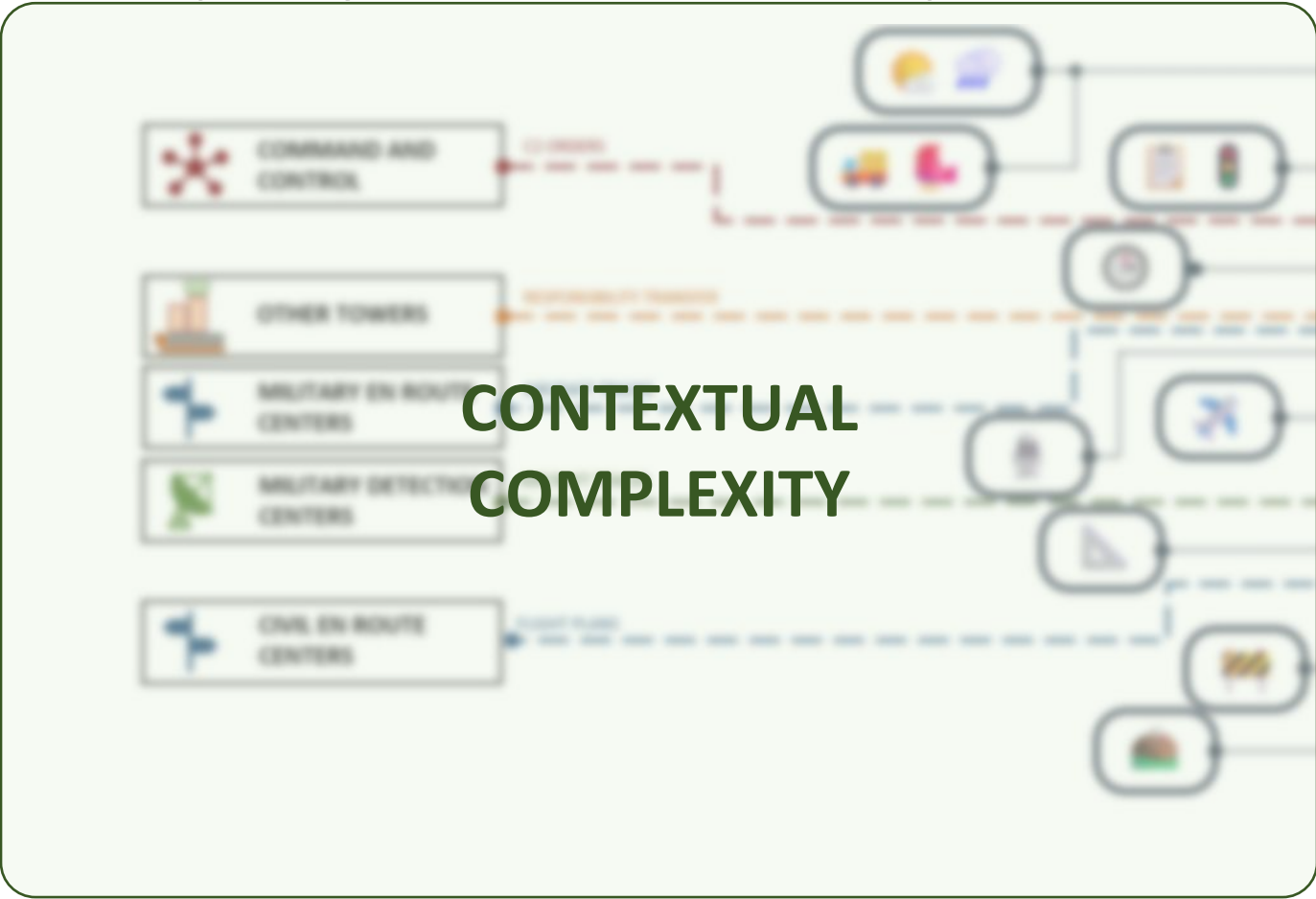
Complexity of air traffic control operations

CONTEXTUAL
COMPLEXITY

INTERNAL
COMPLEXITY

- AC: Arrival Controller
- RA: Radar Controller
- FC: Final Controller
- TO: Tower Officer
- TC: Tower Controller
- GC: Ground Controller
- CO: Officer
- CA: Approach Controller
- DP: Departure Controller
- TR: Transfer Controller
- MO: Mission Officer
- MA: Mission Controller
- CH: High Test Officer
- EX: High Test Controller
- IO: Information Officer
- IS: Information Operator
- BP: Track Controller
- PI: Pilot
- ST: Server Technician
- RT: Radar Technician
- BT: Barrier Technician
- NT: Navigation Technician
- CT: Crash Team
- FB: Fire Brigade
- RT: Rescue Team
- LS: Local Security

Complexity of air traffic control operations



(Design) Complexity

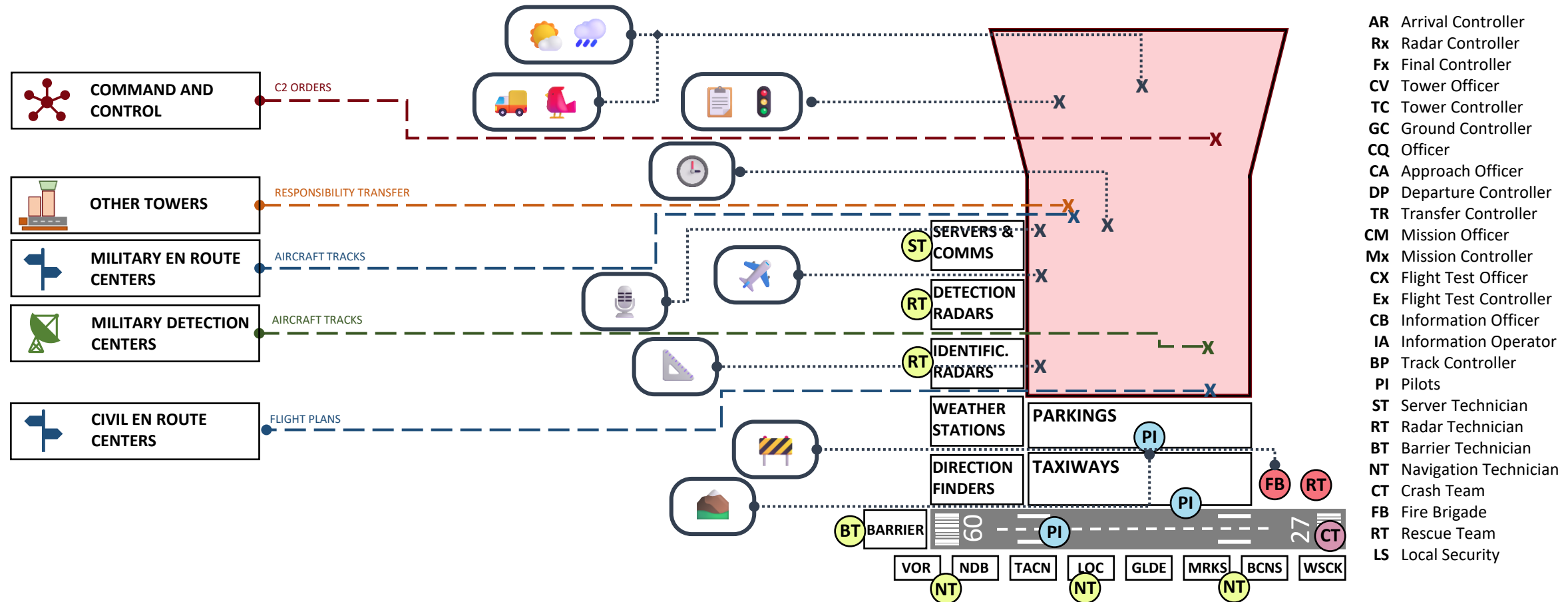
"The measure of how difficult it is to understand how a system will behave or to predict the consequences of changing it."

(Sheard and Mostashari, 2009)

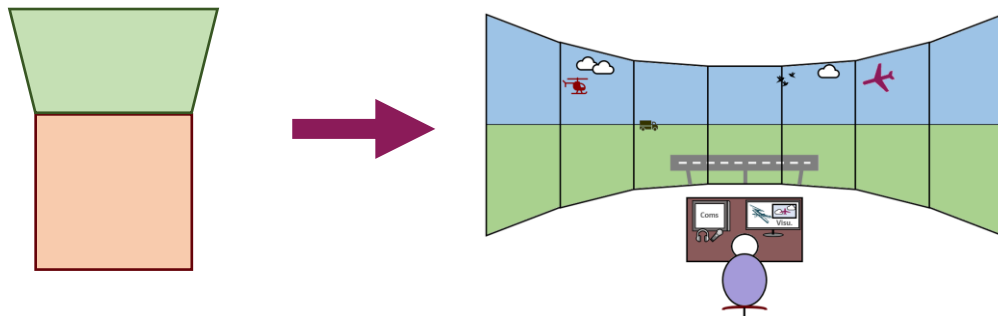
"[Sociotechnical] systems show a level of complexity making design activities very hard."

(Maier, 1998)

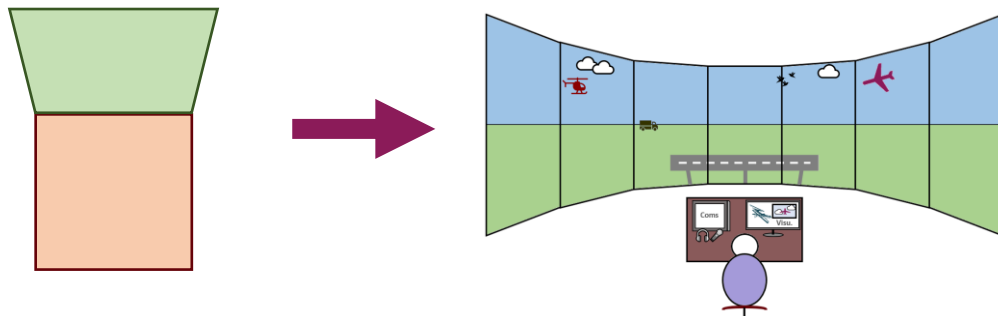
Industrial motivation: context in remote tower design



Industrial motivation: context in remote tower design



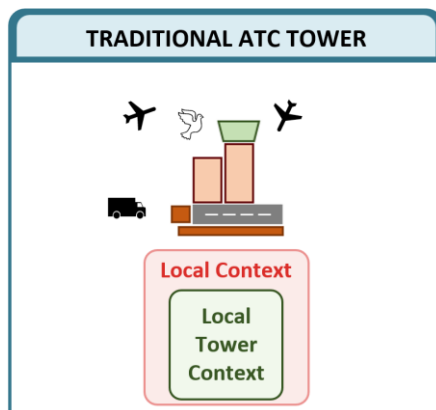
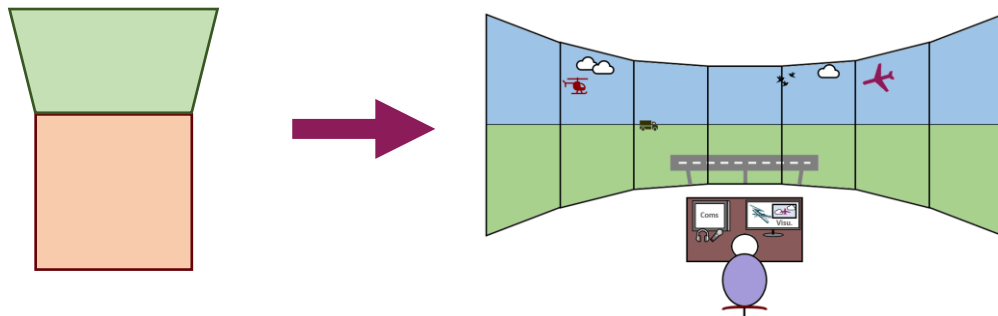
Industrial motivation: context in remote tower design



"The operational context is a key factor to be considered in the definition of the requirements related to remote tower operations."

(EGHD Report, 2017)

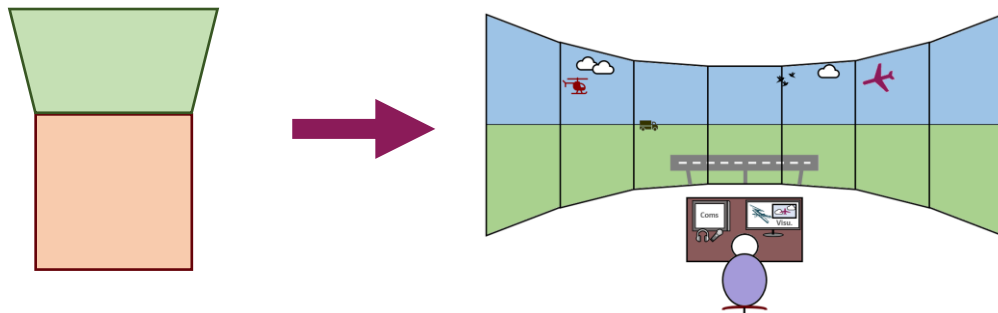
Industrial motivation: context in remote tower design



"The operational context is a key factor to be considered in the definition of the requirements related to remote tower operations."

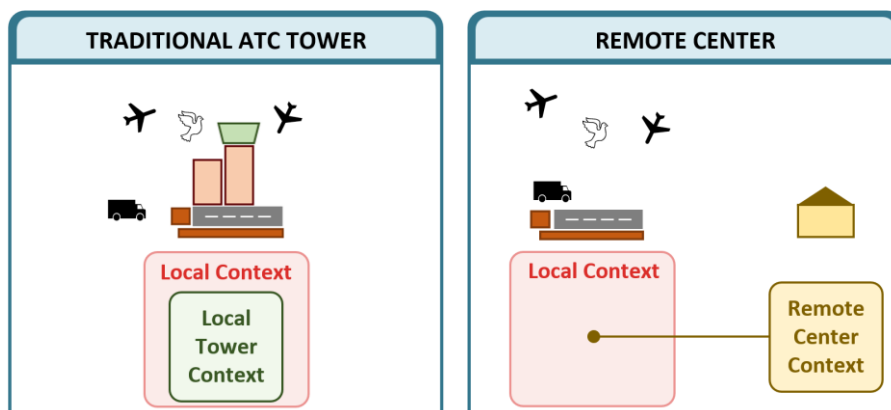
(EGHD Report, 2017)

Industrial motivation: context in remote tower design

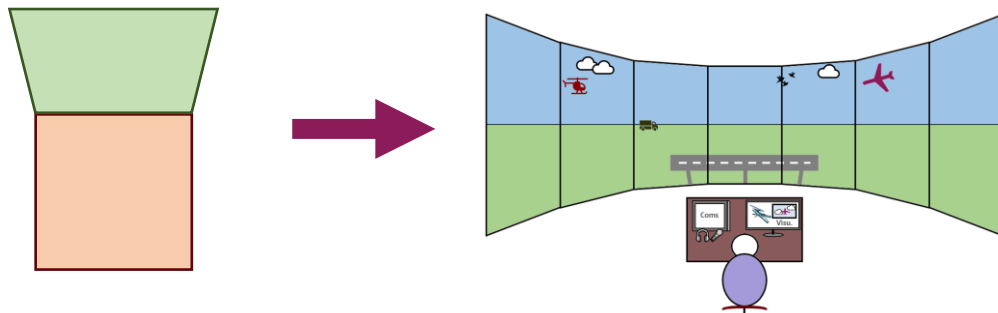


"The operational context is a key factor to be considered in the definition of the requirements related to remote tower operations."

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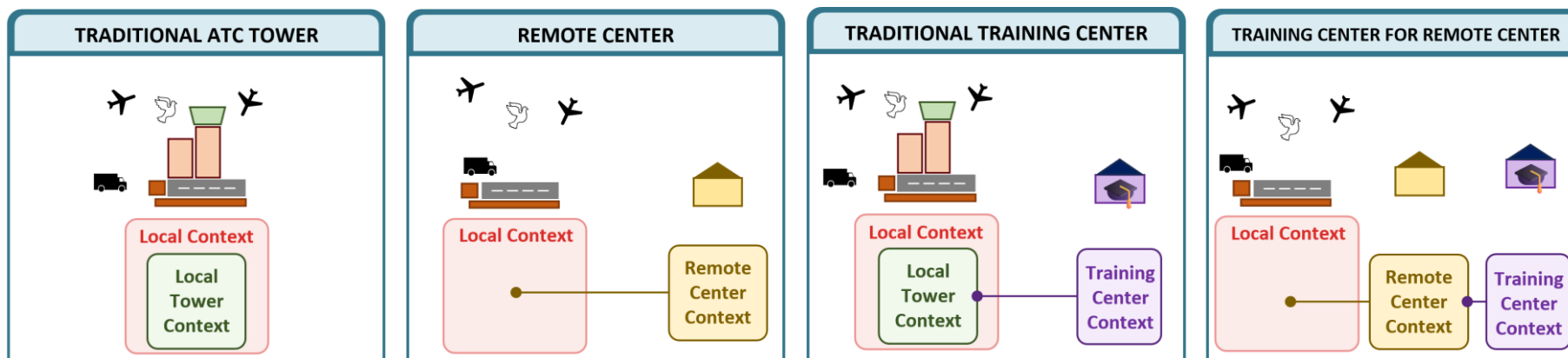


Industrial motivation: context in remote tower design

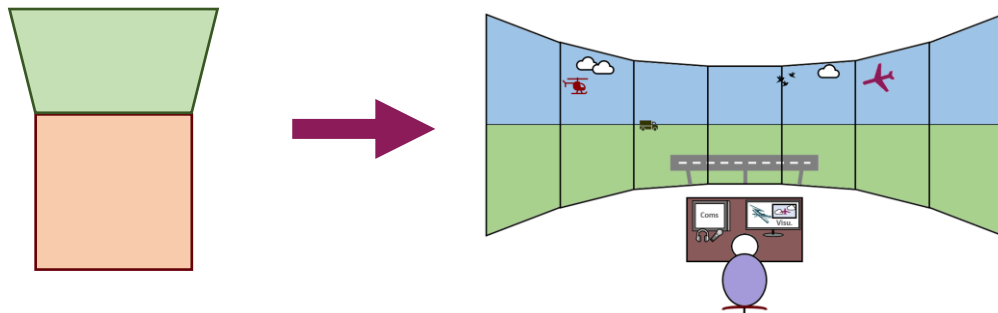


"The operational context is a key factor to be considered in the definition of the requirements related to remote tower operations."

(EGHD Report, 2017)

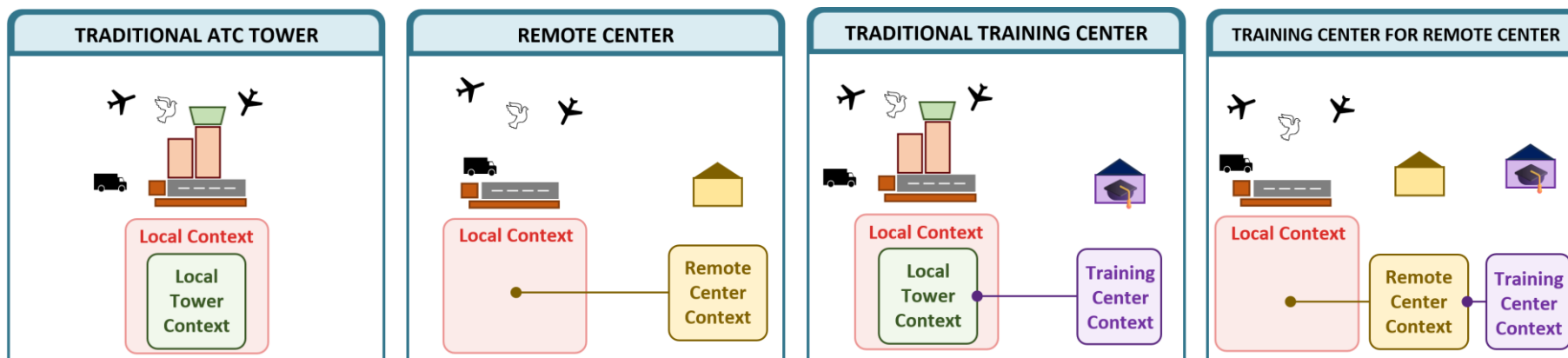


Industrial motivation: context in remote tower design



"The operational context is a key factor to be considered in the definition of the requirements related to remote tower operations."

(EGHD Report, 2017)



Context

"Design of sociotechnical systems can overlook context by focusing solely on technical solutions to the problem."
(Suchman, 2006)

"Identifying which parts of the environment are related to relevant context is not trivial [...]. A method is necessary to study context [...] and determine what is important to take into account during design"
(Engelenburg, 2019)

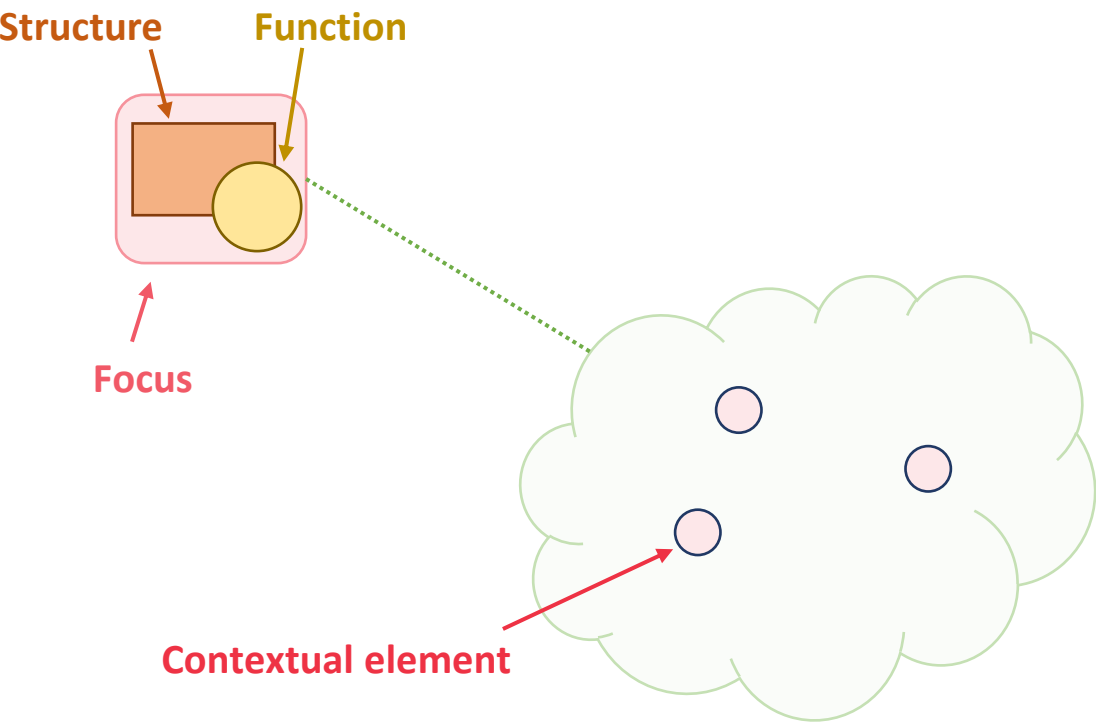
Terminology used in the literature to define the context

Termes utilisés dans les définitions	Notre terminologie
environment(8), not explicit(3), physical environment(2), external(2), surrounding(1), encircling(1), nearby(1), social environment(1), psychological environment(1), outside(1), place(1), emergent(1), where(1) elements(7), information(5), time(4), location(2), knowledge(1), space(1), lighting(1), noise level(1), network connectivity(1), communication costs(1), communication bandwidth(1), social situation(1) relevant(9), of interest(3), subset(2), domain(1), part of the world(1)	relevant contextual elements
user(5), person(2), who(2), actor(2), human(2), agent(2), individual(2), role(1) entity(7), system(7), object(5), product(3), thing(3), something(2), application(2), computer(2), resources(2), equipment(1), hardware(1), software(1), artifact(1), what(1), structures(1), materials(1)	structure
focus(3), specific(3), subject(2), [object] of interest(1), object being processed(1), particular(1), given(1), concept(1), current(1)	focus
behavior(6), task(3), activity(3), operation(3), process(1), execution(1), action(1), computation(1) goal(3), purpose(1), scope(1), service(1), completion(1), mission(1), problem(1)	function
influence(3), constraint(3), be depended upon(2), affect(1), threat(1), enable(1), implications(1), sensitivity(1), induction(1), adaptation(1), changing(1)	influence
help explain(1), solve(1), characterize(1), learn(1), understand(1), classify(1), perceive(1), predict(1), recognize(1), interpret(1), describe(1), reason(1), infer(1), sense(1), meaning(1)	help explain
situation(9), circumstance(5), state(3), underpinnings(2), physical state(1), conceptual state(1), characteristics(1), background(1), setting(1)	situation
conditions(4), factors(2), patterns(2), event(1), happening(1), stimuli(1), causes(1)	event

The context is a historical sequence of situations that influences and helps to understand the behavior of a focus.

Definition of the operational context of STS

The context is a historical sequence of situations that influences and helps to understand the behavior of a focus.



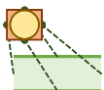
Examples

Type	Elements	Domain
Météorological	Cloudy	{ No Significant Cloud (NSC), Few (FEW), Scattered (SCT), Broken (BKN), Overcast (OVC), Isolate (ISOL) }
	Wind direction	$\mathbb{N} \in [0, 359]$ (degrés)
	QFE (ground pressure)	$\mathbb{N}$ (hPa)
	Dew point	$\mathbb{R}$ (°C)
Aeronautical	Air traffic density	{ Low, Medium, High }
	Presence of birds	Boolean
	Notice to Airmen (NOTAM)	{None, Danger, Theft of important persons, Closed runway, Closed airway, Unusable radio navigation aids, Airspace restrictions, Unusable lights, High obstruction, Obstacle near the aerodrome. }
Cognitive	Mental load	{ 1 – Very low, 2, 3, 4, 5 – Very high }
	Stress level	$\mathbb{N} \in [0, 40]$
...		

The six properties of the operational context of an STS

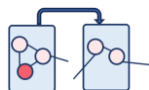
CONTEXT LEVEL

Specific



Context is always relative to some focus object and cannot be discussed in an absolute manner

Transient



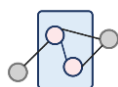
Context is not static, it is a dynamic thing that changes through time

Holistic



Context of a system is more than the sum of the contexts of the parts of the system

Curated



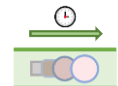
Only a few contextual elements have a real relevance to the system and its behavior

Entangled



Context and systems affect each other's resources and behavior

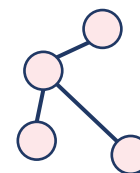
Persistent



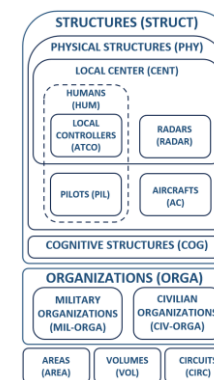
Former values of a contextual element can still have a relevance to the current situation

CONTEXTUAL ELEMENT LEVEL

Contextual Element Typology



ATC Domain Model



Relevance

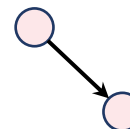
Relevance Intensity

- 5 Absolutely relevant
- 4 Probably relevant
- 3 Rather relevant
- 2 May be relevant
- 1 Not relevant

Relevance Frequency

- 5 Always relevant
- 4 Often relevant
- 3 Several times relevant
- 2 Sometimes relevant
- 1 Never relevant

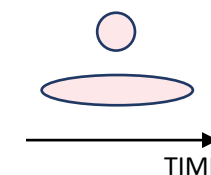
Dependencies



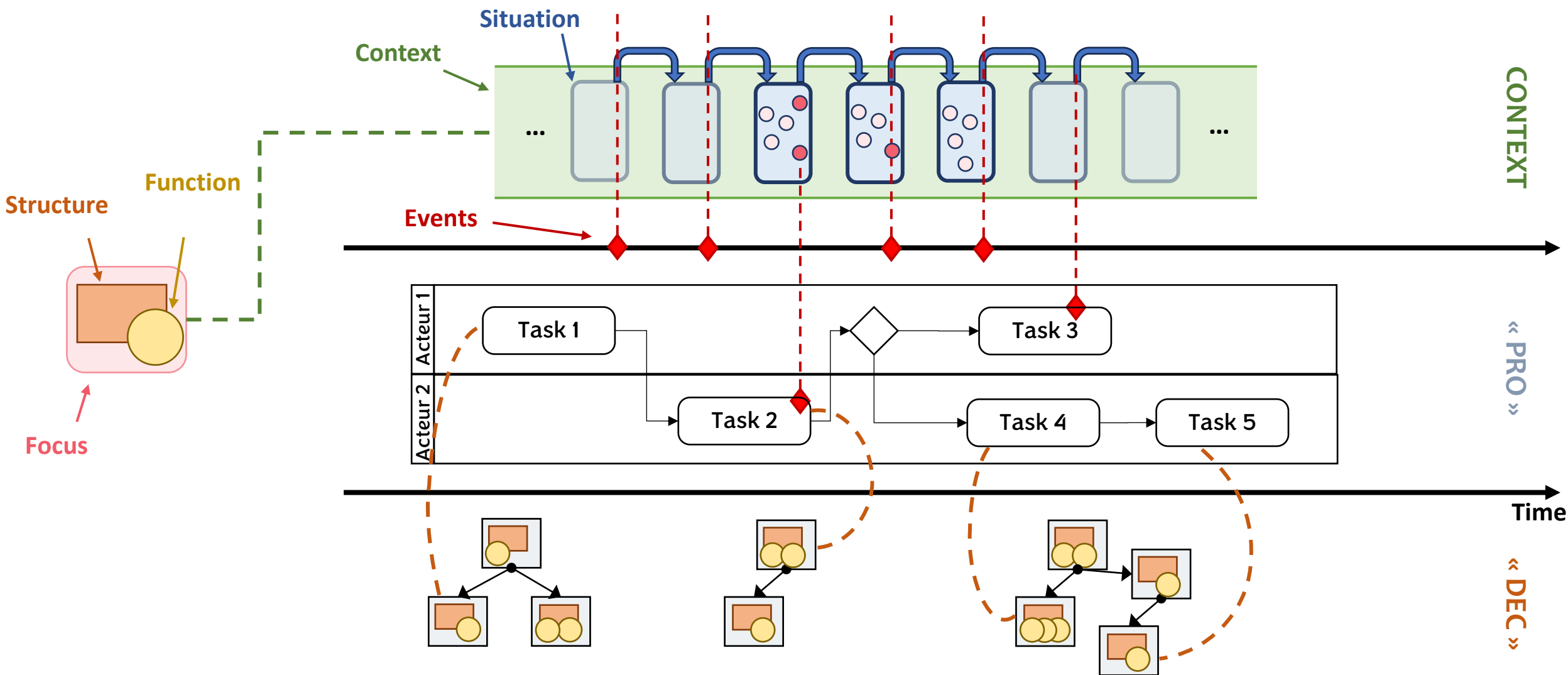
Criticality

Nominal Off-nominal

Temporality



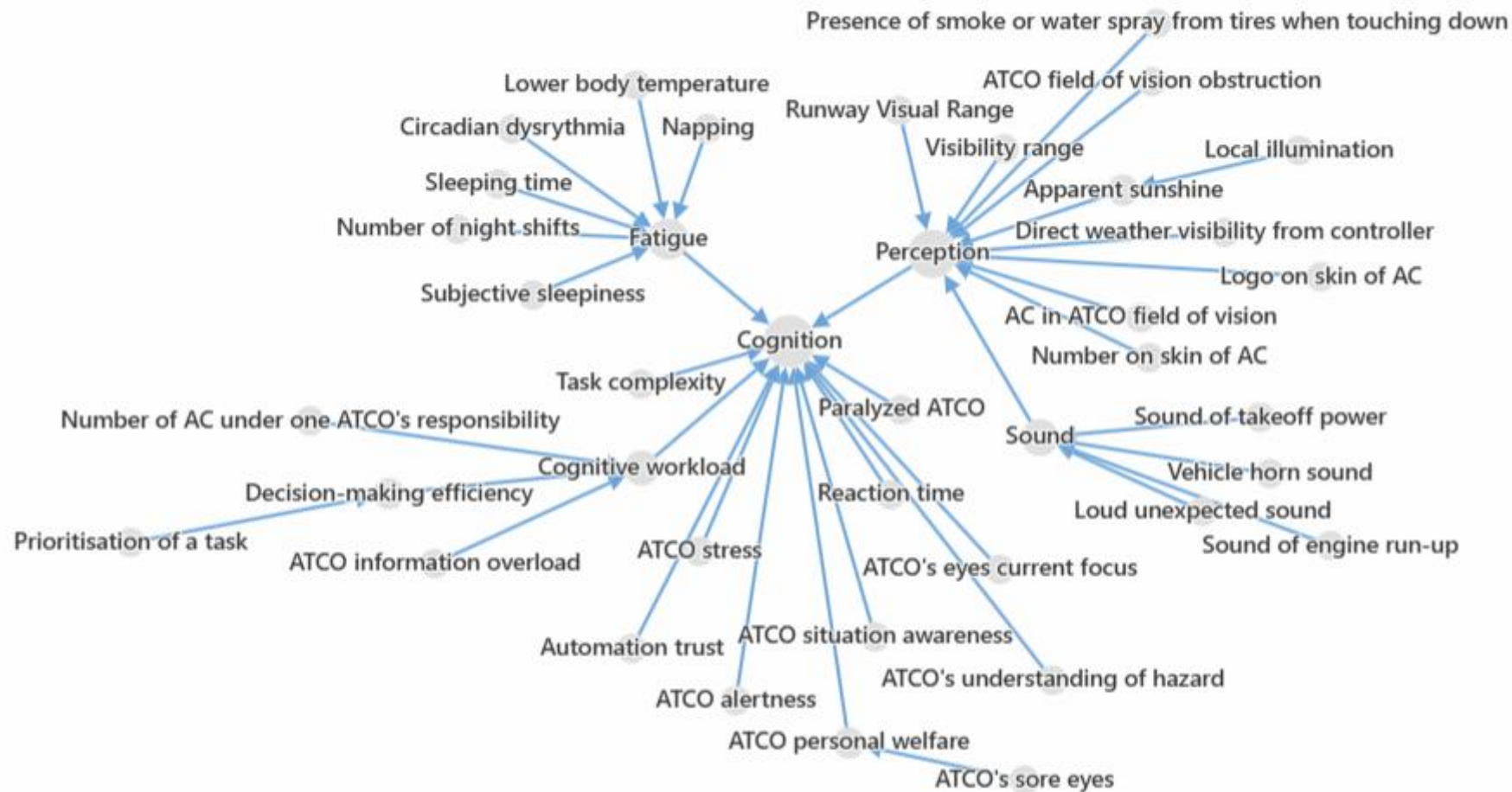
Definition of the operational context of STS: overview



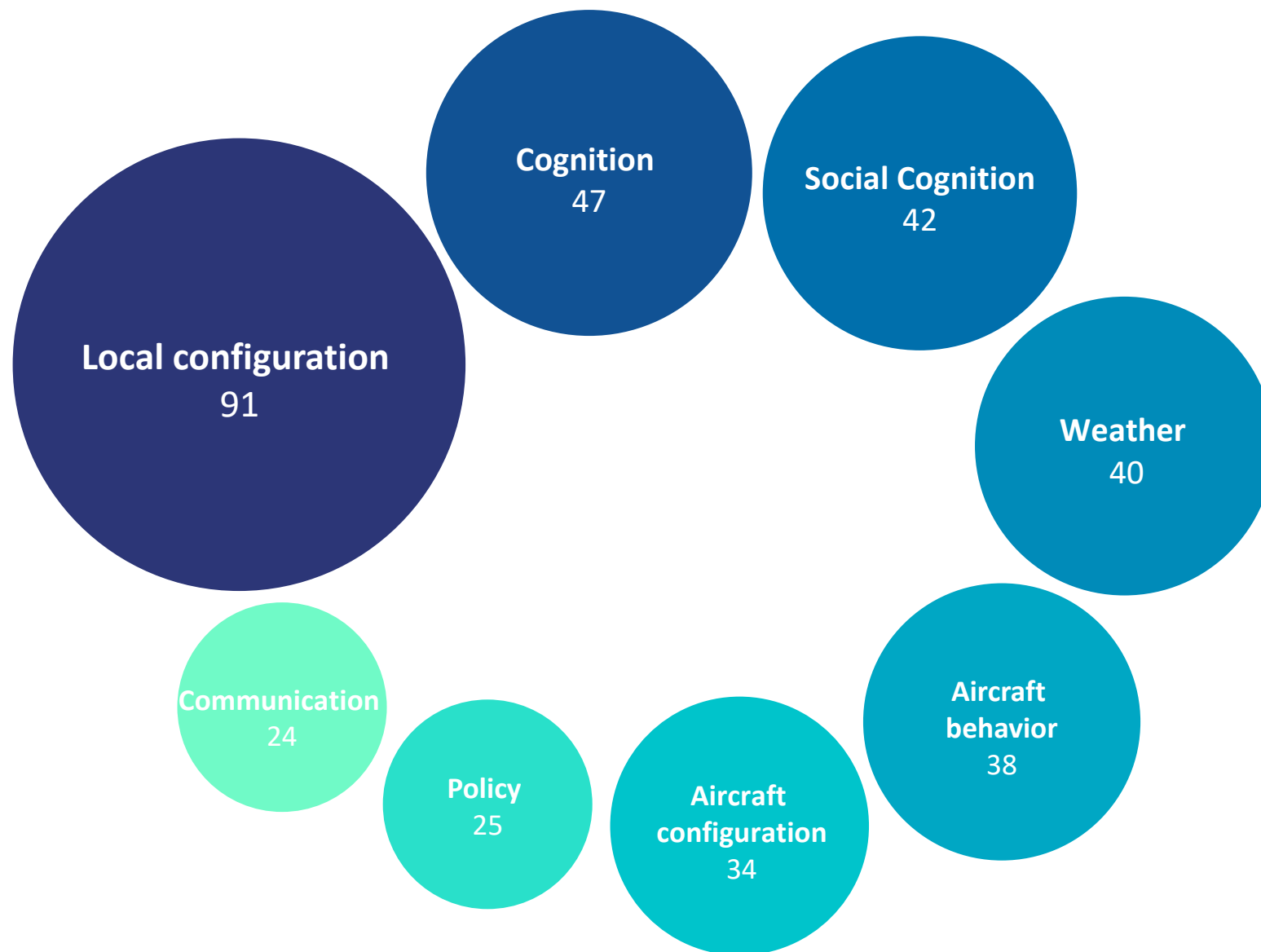
More than 300 contextual elements



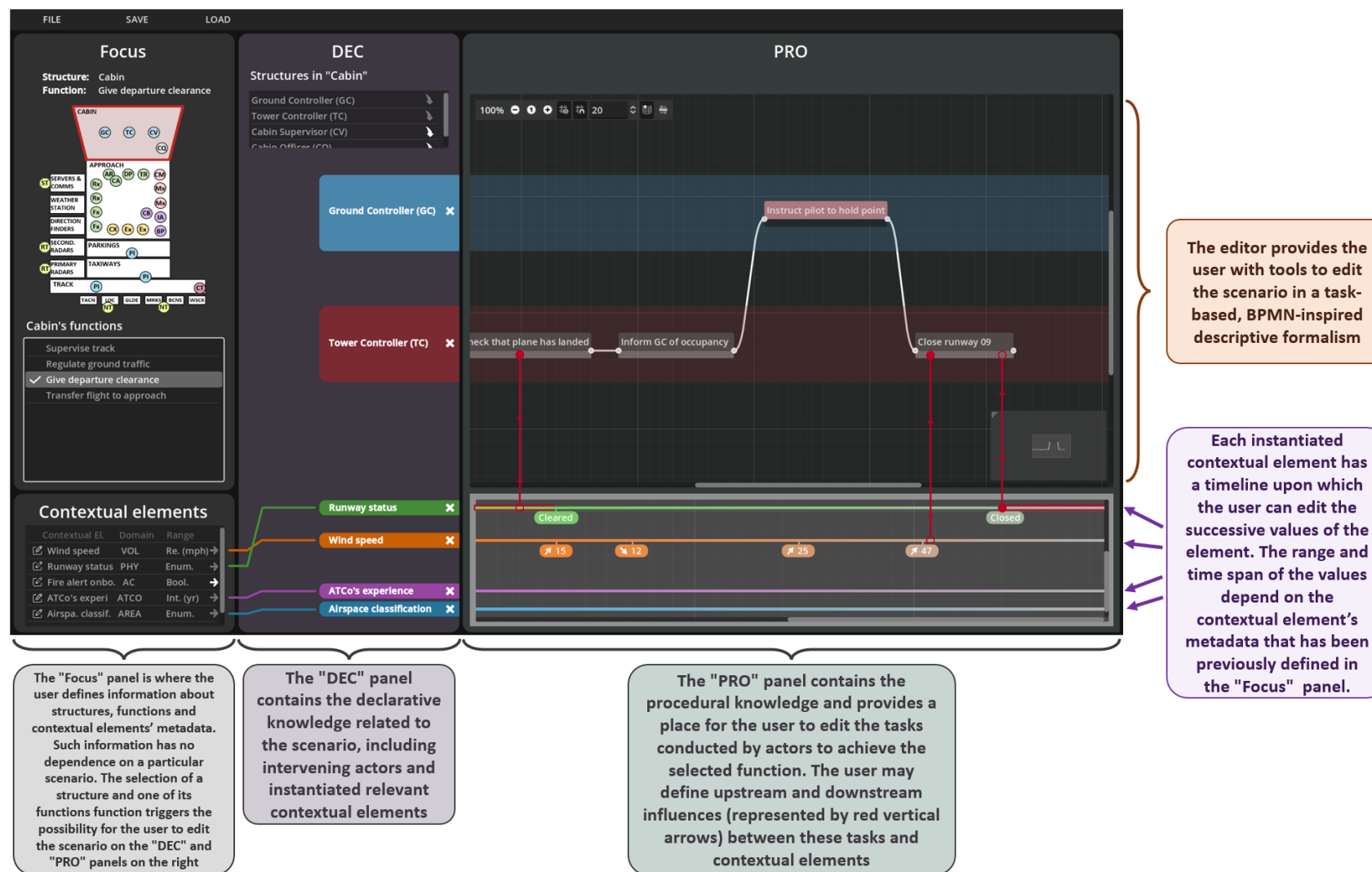
More than 300 contextual elements



There are plenty of potential HSI Domains!



Towards Context-Aware Model-Based HSI



Some closing remarks

HSI is:

- a young discipline (~ 20 years)
- a wide range of multidisciplinary approaches to systems engineering
- rehabilitation of the human and organizational elements in systems engineering
- adapted to complex systems, particularly socio-technical ones

HSI is not:

- incompatible with human-centered methods in systems engineering (human factors, UX, User-Centered Design, agility, etc.)
- a simple methodology: it is also a set of tools and a representation of knowledge
- only related to the design phase of a system

In the long term, we need to:

- develop a theoretical or even mathematical framework for HSI concepts (and systems engineering)
- standardize, structure and equip methodologies so that they can be more easily transferred to industry.
- give greater emphasis to the teaching of HSI (and systems engineering)

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